

Supplementary Material: NetShield

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I. EVIDENCE CLASSIFICATION AND SCOPE

The primary Raspberry Pi campaign contains 12 sessions and 216 accepted arms. A separately frozen cyclic rotating full-admission (CRFA) extension contains 36 accepted arms over the same 12 session/workload identities and three capacities. The primary schedule-anchor reconstruction is derived from accepted primary hardware timestamps; the CRFA timing reconstruction is derived independently from its 9,720 accepted hardware occurrences. Queue-weight replay, phase-randomized control, rotating-frontier software conformance, and multi- n scaling are read-only offline analyses and do not modify either hardware evidence chain.

TABLE I
EVIDENCE CLASSES.

Component	Evidence type	Permitted interpretation
Primary campaign	Hardware	Three-Pi A–F measurements
CRFA extension	Hardware	Three-Pi rotating-frontier measurement
Primary full-raw audit	Raw reconstruction	Accepted-arm traceability
Queue-weight sweep	Offline replay	Frozen-selector sensitivity
Phase randomization	Software sensitivity	Desynchronization counterfactual
Primary schedule anchors	Hardware timestamp reconstruction	Post vs. nominal timing
CRFA timing reconstruction	Hardware timestamp reconstruction	Frontier attenuation/timing
Probe-count study	Discrete-event conformance	Analytical boundary check
Rotating construction	Discrete-event conformance	Ideal byte-frontier construction

II. FROZEN ACTION TABLE AND I-QPG THRESHOLDS

TABLE II
FROZEN ACTION LATTICE AND $v = 1$ THRESHOLDS AGAINST H-NONE.

Action	Payload (B)	Utility	Threshold Q (B)
L-NONE	0	0.758501	–
L-ALERT	256	0.825168	221.84
L-SUMMARY	4096	0.891835	30.14
M-NONE	0	0.766968	–
M-ALERT	256	0.833635	254.91
M-SUMMARY	8192	0.900302	16.10
H-NONE	0	0.768378	–
H-ALERT	256	0.835044	260.41
H-SUMMARY	16384	0.901711	8.14
H-FULL	131072	0.968378	1.53

III. THEOREM DERIVATIONS

For m remotely represented probes and k full-evidence probes, payload is $mS + k(B - S)$. Hence one-round feasibility is

$$kB + (m - k)S \leq C,$$

and, for $C \geq mS$,

$$k^*(C, m) = \min \left\{ m, \left\lfloor \frac{C - mS}{B - S} \right\rfloor \right\}.$$

With $\rho = C/(nB)$, $\sigma = S/B$, $r = m/n$, and $f = k/n$, the continuous frontier is $\sigma r + (1 - \sigma)f \leq \rho$.

If non-selected probes may send zero remote payload, payload for k full requests is minimized by setting $m = k$, so the unrestricted full-evidence byte frontier is

$$K_{\text{full}}^{\max} = \min\{n, \lfloor C/B \rfloor\}.$$

Provided $C \geq nS$, requiring all probes to remain remotely represented gives

$$K_{\text{all}}^{\max} = \left[\left\lfloor \frac{C - nS}{B - S} \right\rfloor \right]_0^n.$$

If $C < nS$, the all-summary baseline is itself infeasible; clipping the count to zero does not make universal coverage feasible. For the formal $n = 3$, $B = 128$ KiB, $S = 16$ KiB system, unrestricted versus all-probe counts are $(1, 0)$, $(2, 1)$, and $(3, 3)$ at low, mid, and high capacity. Thus universal per-round coverage costs one full-evidence slot at low and mid capacity.

For total admitted bytes A_t released at the start of round t , an idealized preemptive fluid server with common relative deadline D rounds satisfies the demand condition

$$\sum_{\tau=s}^t A_\tau \leq C(t - s + D), \quad s \leq t.$$

$D = 1$ reduces to per-round feasibility. Dividing the $s = 1$ case by T gives the long-run load condition

$$\limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T A_t \leq C.$$

The latter is a load-admissibility condition, not a deterministic bounded-delay guarantee for arbitrary burst processes.

For synchronized fixed-full overload from empty queues with $nB > C$, work-conserving service removes $\min\{Q_\Sigma(t) + nB, C\} = C$ per round, giving $Q_\Sigma(T) = T(nB - C)$. At mid capacity, the post-full per-probe queue is $B/3 = 43,690.6$ B, or approximately 42.7 KiB. This exceeds every positive-payload threshold; the next H-NONE round drains the residual, producing the frozen I-QPG full/none alternation.

In the reduced H-FULL/H-NONE desynchronized regime, if aggregate backlog is rate-stable and the server is asymptotically saturated, flow conservation gives

$$p^* = \min(1, C/(nB)) = \min(1, \rho).$$

The phase-randomized software study matches $1/3$, $2/3$, and 1 at the three capacities.

Finally, with $K = K_{\text{full}}^{\max}$, selecting probes $\mathcal{S}_t = \{(tK + j) \bmod n : j = 0, \dots, K - 1\}$ attains the unrestricted byte frontier every round. Its period is $n/\gcd(n, K)$, and each probe is selected $K/\gcd(n, K)$ times per period.

IV. ANALYTICAL EXTENSIONS FOR DEPLOYMENT

These extensions derive planning consequences of the one-round byte model. They are not new hardware measurements, fitted arrival models, or guarantees of physical deadline attainment. MB denotes 10^6 bytes; KiB and MiB denote 2^{10} and 2^{20} bytes. The integer boundaries remain the operational reference for the three-probe measurements.

A. Probabilistic Requests and Capacity Planning

Assume empty incoming queues, fixed $0 < S < B$, and a budget $C \geq nS$ that can first reserve summaries for all n probes. Within a round, each probe independently requests a full upload with common probability $p \in [0, 1]$; otherwise it requests a summary. Independence across rounds is not needed for the following marginal per-round calculation. Let

$$J \sim \text{Binomial}(n, p), \quad h(C) = \min \left\{ n, \left\lfloor \frac{C - nS}{B - S} \right\rfloor \right\}. \quad (1)$$

Under a count-maximizing summary-preserving admission rule, the admitted full count is $J_{\text{adm}} = \min\{J, h(C)\}$. Consequently,

$$P_{\text{deg}}(C) = \Pr\{J > h(C)\} = \sum_{j=h(C)+1}^n \binom{n}{j} p^j (1-p)^{n-j}, \quad (2)$$

$$\mathbb{E}[J_{\text{adm}}] = \sum_{j=0}^n \min\{j, h(C)\} \binom{n}{j} p^j (1-p)^{n-j}. \quad (3)$$

Here P_{deg} is the probability that a round needs at least one full-to-summary downgrade, not a per-request loss probability. The expected number of downgraded requests is $np - \mathbb{E}[J_{\text{adm}}]$. Non-requesting and downgraded probes still retain summary evidence. Both equations concern admitted bytes, not returned core results.

For a target $\epsilon \in [0, 1]$, define the upper-tail quantile

$$q_\epsilon = \min\{q \in \{0, \dots, n\} : \Pr(J > q) \leq \epsilon\}. \quad (4)$$

The minimum budget within this model that preserves all summaries and satisfies $P_{\text{deg}} \leq \epsilon$ is

$$\boxed{C_\epsilon = nS + (B - S)q_\epsilon}. \quad (5)$$

Indeed, $h(C_\epsilon) = q_\epsilon$. Any smaller budget either fails the summary baseline (when $q_\epsilon = 0$) or admits fewer than q_ϵ full requests, contradicting quantile minimality. Since $h(C)$ is non-decreasing, expected admitted richness is also non-decreasing in C .

a) *Illustrative calculation, not an empirical result:* With the paper's $n = 3$, $B = 131,072$ B and $S = 16,384$ B, suppose independently $p = 2/3$ and choose $\epsilon = 1/3$. Then $\Pr(J > 1) = 20/27$ but $\Pr(J > 2) = 8/27$, giving $q_\epsilon = 2$ and $C_\epsilon = 278,528$ B (272 KiB). At this budget, $\mathbb{E}[J_{\text{adm}}] = 46/27$ and the expected downgrade count is $8/27$ per round. The planning budget differs from the hardware mid budget of 256 KiB because it reserves a summary for the third probe. The numerical choice $p = 2/3$ does not establish independence of the QPG traces or identify this planning rule with CRFA, which assigns H-NONE to non-selected probes.

For a large- n interpretation, hold B, S, p and normalized capacity $\rho = C_n/(nB)$ fixed, with $\rho \geq \sigma = S/B$. Then

$$\frac{\mathbb{E}[J_{\text{adm}}]}{n} \rightarrow \min \left\{ p, \min \left(1, \frac{\rho - \sigma}{1 - \sigma} \right) \right\}. \quad (6)$$

This follows because $J/n \rightarrow p$ in probability, $h(C_n)/n$ converges to the clipped capacity term, and the normalized admitted count is bounded by one. For $\rho < \sigma$, universal summary coverage is infeasible rather than a zero-richness feasible point. With correlated within-round requests, the binomial tail is not justified; the same quantile construction requires the actual joint request-count distribution. Queueing, security utility, and physical timing remain outside this planning calculation.

B. Heterogeneous Payloads: Summary Baseline and Upgrade Costs

Fix the set $\mathcal{M}$ of probes required to retain remote evidence, with $0 < S_i < B_i$ for every $i \in \mathcal{M}$. Let

$$S_{\mathcal{M}} = \sum_{i \in \mathcal{M}} S_i, \quad d_i = B_i - S_i > 0, \quad R = C - S_{\mathcal{M}}. \quad (7)$$

If $R < 0$, the prescribed coverage set is infeasible. Otherwise, upgrading $\mathcal{F} \subseteq \mathcal{M}$ to full evidence is feasible exactly when

$$S_{\mathcal{M}} + \sum_{i \in \mathcal{F}} d_i \leq C. \quad (8)$$

For a pure full-count objective (equal positive value per upgrade), sort $d_{(1)} \leq \dots \leq d_{(m)}$, where $m = |\mathcal{M}|$. The optimum is

$$k_{\text{het}}^* = \max \left\{ k \in \{0, \dots, m\} : \sum_{j=1}^k d_{(j)} \leq R \right\}. \quad (9)$$

Any set of k upgrades costs at least the sum of the k smallest costs, which proves feasibility and optimality of this count rule. When payloads are homogeneous, it reduces to $\min\{m, \lfloor (C - mS)/(B - S) \rfloor\}$ for $C \geq mS$.

For general nonnegative values w_i , the formulation is instead

$$\max_{x_i \in \{0,1\}} \sum_{i \in \mathcal{M}} w_i x_i \quad \text{subject to} \quad \sum_{i \in \mathcal{M}} d_i x_i \leq R, \quad (10)$$

a 0/1 knapsack problem. Equal value is not equal value density. With $R = 3$ and cost/value pairs $(2, 2)$ and $(3, 3)$, both densities equal one, but choosing the cheaper upgrade yields value 2 while choosing the other yields value 3. Thus equal density does not justify an increasing-cost greedy rule for the weighted problem. No heterogeneous-payload hardware validation or security-value optimum is claimed here; the scope is a fixed coverage set with separable values and no additional fairness constraints.

C. Continuous Exchange Rate versus Discrete Feasibility

In the continuous relaxation, $0 \leq f \leq r \leq 1$ and $\sigma r + (1 - \sigma)f \leq \rho$. Along any non-degenerate feasible segment of the active budget line,

$$\frac{dr}{df} = -\frac{1 - \sigma}{\sigma}. \quad (11)$$

This is a local coverage–richness exchange rate, not an integer scheduling rule. With $\sigma = 1/8$, its magnitude is seven. The full-coverage boundary point $r = 1$, $f = (\rho - \sigma)/(1 - \sigma)$ applies only for $\sigma \leq \rho \leq 1$; for $\rho > 1$, $(r, f) = (1, 1)$ is already maximal although the byte budget is slack.

Discrete Pareto points can likewise leave unused capacity. At $n = 3$ and $C = B$ with $S = B/8$, the all-summary allocation $(m, k) = (3, 0)$ costs only $3S = 3B/8$. It nevertheless cannot increase k while retaining $m = 3$, since one full upload plus two summaries costs $B + 2S > B$. It is therefore undominated in the integer coverage/count ordering despite its slack byte budget. For the hardware testbed, the integer slot price and CRFA selection schedule, rather than a continuous Pareto assertion, govern the feasible operating points.

V. COMPLETE QUEUE-WEIGHT SWEEP RESULTS

The $v = 1$ identity gate covers 36 accepted physical D cells and four deterministic checks per cell. Low-capacity cycle/action composition changes with v ; no all- v cycle-identity claim is made.

TABLE III: Complete frozen-selector v -sweep results (432 retained rows).

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S01	1	low	0.244444	9191424	87382	4
F12_S02	1	low	0.244444	9191424	87382	4
F12_S03	1	low	0.244444	9191424	87382	4
F12_S04	1	low	0.244444	9191424	87382	4
F12_S05	1	low	0.244444	9191424	87382	4
F12_S06	1	low	0.244444	9191424	87382	4
F12_S07	1	low	0.244444	9191424	87382	4
F12_S08	1	low	0.244444	9191424	87382	4
F12_S09	1	low	0.244444	9191424	87382	4
F12_S10	1	low	0.244444	9191424	87382	4
F12_S11	1	low	0.244444	9191424	87382	4
F12_S12	1	low	0.244444	9191424	87382	4
F12_S01	1	mid	0.500000	17694720	43691	2
F12_S02	1	mid	0.500000	17694720	43691	2
F12_S03	1	mid	0.500000	17694720	43691	2
F12_S04	1	mid	0.500000	17694720	43691	2
F12_S05	1	mid	0.500000	17694720	43691	2
F12_S06	1	mid	0.500000	17694720	43691	2
F12_S07	1	mid	0.500000	17694720	43691	2
F12_S08	1	mid	0.500000	17694720	43691	2
F12_S09	1	mid	0.500000	17694720	43691	2
F12_S10	1	mid	0.500000	17694720	43691	2
F12_S11	1	mid	0.500000	17694720	43691	2
F12_S12	1	mid	0.500000	17694720	43691	2
F12_S01	1	high	1.000000	35389440	0	1
F12_S02	1	high	1.000000	35389440	0	1
F12_S03	1	high	1.000000	35389440	0	1
F12_S04	1	high	1.000000	35389440	0	1
F12_S05	1	high	1.000000	35389440	0	1
F12_S06	1	high	1.000000	35389440	0	1
F12_S07	1	high	1.000000	35389440	0	1
F12_S08	1	high	1.000000	35389440	0	1
F12_S09	1	high	1.000000	35389440	0	1
F12_S10	1	high	1.000000	35389440	0	1
F12_S11	1	high	1.000000	35389440	0	1
F12_S12	1	high	1.000000	35389440	0	1
F12_S01	3	low	0.244444	9191424	87382	4
F12_S02	3	low	0.244444	9191424	87382	4
F12_S03	3	low	0.244444	9191424	87382	4
F12_S04	3	low	0.244444	9191424	87382	4
F12_S05	3	low	0.244444	9191424	87382	4
F12_S06	3	low	0.244444	9191424	87382	4
F12_S07	3	low	0.244444	9191424	87382	4
F12_S08	3	low	0.244444	9191424	87382	4
F12_S09	3	low	0.244444	9191424	87382	4
F12_S10	3	low	0.244444	9191424	87382	4
F12_S11	3	low	0.244444	9191424	87382	4
F12_S12	3	low	0.244444	9191424	87382	4
F12_S01	3	mid	0.500000	17694720	43691	2
F12_S02	3	mid	0.500000	17694720	43691	2
F12_S03	3	mid	0.500000	17694720	43691	2
F12_S04	3	mid	0.500000	17694720	43691	2
F12_S05	3	mid	0.500000	17694720	43691	2
F12_S06	3	mid	0.500000	17694720	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S07	3	mid	0.500000	17694720	43691	2
F12_S08	3	mid	0.500000	17694720	43691	2
F12_S09	3	mid	0.500000	17694720	43691	2
F12_S10	3	mid	0.500000	17694720	43691	2
F12_S11	3	mid	0.500000	17694720	43691	2
F12_S12	3	mid	0.500000	17694720	43691	2
F12_S01	3	high	1.000000	35389440	0	1
F12_S02	3	high	1.000000	35389440	0	1
F12_S03	3	high	1.000000	35389440	0	1
F12_S04	3	high	1.000000	35389440	0	1
F12_S05	3	high	1.000000	35389440	0	1
F12_S06	3	high	1.000000	35389440	0	1
F12_S07	3	high	1.000000	35389440	0	1
F12_S08	3	high	1.000000	35389440	0	1
F12_S09	3	high	1.000000	35389440	0	1
F12_S10	3	high	1.000000	35389440	0	1
F12_S11	3	high	1.000000	35389440	0	1
F12_S12	3	high	1.000000	35389440	0	1
F12_S01	10	low	0.300000	10838016	87386	10
F12_S02	10	low	0.300000	10838016	87386	10
F12_S03	10	low	0.300000	10838016	87386	10
F12_S04	10	low	0.300000	10838016	87386	10
F12_S05	10	low	0.300000	10838016	87386	10
F12_S06	10	low	0.300000	10838016	87386	10
F12_S07	10	low	0.300000	10838016	87386	10
F12_S08	10	low	0.300000	10838016	87386	10
F12_S09	10	low	0.300000	10838016	87386	10
F12_S10	10	low	0.300000	10838016	87386	10
F12_S11	10	low	0.300000	10838016	87386	10
F12_S12	10	low	0.300000	10838016	87386	10
F12_S01	10	mid	0.500000	17694720	43691	2
F12_S02	10	mid	0.500000	17694720	43691	2
F12_S03	10	mid	0.500000	17694720	43691	2
F12_S04	10	mid	0.500000	17694720	43691	2
F12_S05	10	mid	0.500000	17694720	43691	2
F12_S06	10	mid	0.500000	17694720	43691	2
F12_S07	10	mid	0.500000	17694720	43691	2
F12_S08	10	mid	0.500000	17694720	43691	2
F12_S09	10	mid	0.500000	17694720	43691	2
F12_S10	10	mid	0.500000	17694720	43691	2
F12_S11	10	mid	0.500000	17694720	43691	2
F12_S12	10	mid	0.500000	17694720	43691	2
F12_S01	10	high	1.000000	35389440	0	1
F12_S02	10	high	1.000000	35389440	0	1
F12_S03	10	high	1.000000	35389440	0	1
F12_S04	10	high	1.000000	35389440	0	1
F12_S05	10	high	1.000000	35389440	0	1
F12_S06	10	high	1.000000	35389440	0	1
F12_S07	10	high	1.000000	35389440	0	1
F12_S08	10	high	1.000000	35389440	0	1
F12_S09	10	high	1.000000	35389440	0	1
F12_S10	10	high	1.000000	35389440	0	1
F12_S11	10	high	1.000000	35389440	0	1
F12_S12	10	high	1.000000	35389440	0	1
F12_S01	30	low	0.322222	11476992	87398	28
F12_S02	30	low	0.322222	11476992	87398	28
F12_S03	30	low	0.322222	11476992	87398	28
F12_S04	30	low	0.322222	11476992	87398	28
F12_S05	30	low	0.322222	11476992	87398	28
F12_S06	30	low	0.322222	11476992	87398	28
F12_S07	30	low	0.322222	11476992	87398	28
F12_S08	30	low	0.322222	11476992	87398	28

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S09	30	low	0.322222	11476992	87398	28
F12_S10	30	low	0.322222	11476992	87398	28
F12_S11	30	low	0.322222	11476992	87398	28
F12_S12	30	low	0.322222	11476992	87398	28
F12_S01	30	mid	0.500000	17694720	43691	2
F12_S02	30	mid	0.500000	17694720	43691	2
F12_S03	30	mid	0.500000	17694720	43691	2
F12_S04	30	mid	0.500000	17694720	43691	2
F12_S05	30	mid	0.500000	17694720	43691	2
F12_S06	30	mid	0.500000	17694720	43691	2
F12_S07	30	mid	0.500000	17694720	43691	2
F12_S08	30	mid	0.500000	17694720	43691	2
F12_S09	30	mid	0.500000	17694720	43691	2
F12_S10	30	mid	0.500000	17694720	43691	2
F12_S11	30	mid	0.500000	17694720	43691	2
F12_S12	30	mid	0.500000	17694720	43691	2
F12_S01	30	high	1.000000	35389440	0	1
F12_S02	30	high	1.000000	35389440	0	1
F12_S03	30	high	1.000000	35389440	0	1
F12_S04	30	high	1.000000	35389440	0	1
F12_S05	30	high	1.000000	35389440	0	1
F12_S06	30	high	1.000000	35389440	0	1
F12_S07	30	high	1.000000	35389440	0	1
F12_S08	30	high	1.000000	35389440	0	1
F12_S09	30	high	1.000000	35389440	0	1
F12_S10	30	high	1.000000	35389440	0	1
F12_S11	30	high	1.000000	35389440	0	1
F12_S12	30	high	1.000000	35389440	0	1
F12_S01	100	low	0.333333	11821056	87436	—
F12_S02	100	low	0.333333	11821056	87436	—
F12_S03	100	low	0.333333	11821056	87436	—
F12_S04	100	low	0.333333	11821056	87436	—
F12_S05	100	low	0.333333	11821056	87436	—
F12_S06	100	low	0.333333	11821056	87436	—
F12_S07	100	low	0.333333	11821056	87436	—
F12_S08	100	low	0.333333	11821056	87436	—
F12_S09	100	low	0.333333	11821056	87436	—
F12_S10	100	low	0.333333	11821056	87436	—
F12_S11	100	low	0.333333	11821056	87436	—
F12_S12	100	low	0.333333	11821056	87436	—
F12_S01	100	mid	0.500000	17694720	43691	2
F12_S02	100	mid	0.500000	17694720	43691	2
F12_S03	100	mid	0.500000	17694720	43691	2
F12_S04	100	mid	0.500000	17694720	43691	2
F12_S05	100	mid	0.500000	17694720	43691	2
F12_S06	100	mid	0.500000	17694720	43691	2
F12_S07	100	mid	0.500000	17694720	43691	2
F12_S08	100	mid	0.500000	17694720	43691	2
F12_S09	100	mid	0.500000	17694720	43691	2
F12_S10	100	mid	0.500000	17694720	43691	2
F12_S11	100	mid	0.500000	17694720	43691	2
F12_S12	100	mid	0.500000	17694720	43691	2
F12_S01	100	high	1.000000	35389440	0	1
F12_S02	100	high	1.000000	35389440	0	1
F12_S03	100	high	1.000000	35389440	0	1
F12_S04	100	high	1.000000	35389440	0	1
F12_S05	100	high	1.000000	35389440	0	1
F12_S06	100	high	1.000000	35389440	0	1
F12_S07	100	high	1.000000	35389440	0	1
F12_S08	100	high	1.000000	35389440	0	1
F12_S09	100	high	1.000000	35389440	0	1
F12_S10	100	high	1.000000	35389440	0	1

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S11	100	high	1.000000	35389440	0	1
F12_S12	100	high	1.000000	35389440	0	1
F12_S01	168	low	0.244444	9209088	87382	4
F12_S02	168	low	0.244444	9209088	87382	4
F12_S03	168	low	0.244444	9209088	87382	4
F12_S04	168	low	0.244444	9209088	87382	4
F12_S05	168	low	0.244444	9209088	87382	4
F12_S06	168	low	0.244444	9209088	87382	4
F12_S07	168	low	0.244444	9209088	87382	4
F12_S08	168	low	0.244444	9209088	87382	4
F12_S09	168	low	0.244444	9209088	87382	4
F12_S10	168	low	0.244444	9209088	87382	4
F12_S11	168	low	0.244444	9209088	87382	4
F12_S12	168	low	0.244444	9209088	87382	4
F12_S01	168	mid	0.500000	17729280	43691	2
F12_S02	168	mid	0.500000	17729280	43691	2
F12_S03	168	mid	0.500000	17729280	43691	2
F12_S04	168	mid	0.500000	17729280	43691	2
F12_S05	168	mid	0.500000	17729280	43691	2
F12_S06	168	mid	0.500000	17729280	43691	2
F12_S07	168	mid	0.500000	17729280	43691	2
F12_S08	168	mid	0.500000	17729280	43691	2
F12_S09	168	mid	0.500000	17729280	43691	2
F12_S10	168	mid	0.500000	17729280	43691	2
F12_S11	168	mid	0.500000	17729280	43691	2
F12_S12	168	mid	0.500000	17729280	43691	2
F12_S01	168	high	1.000000	35389440	0	1
F12_S02	168	high	1.000000	35389440	0	1
F12_S03	168	high	1.000000	35389440	0	1
F12_S04	168	high	1.000000	35389440	0	1
F12_S05	168	high	1.000000	35389440	0	1
F12_S06	168	high	1.000000	35389440	0	1
F12_S07	168	high	1.000000	35389440	0	1
F12_S08	168	high	1.000000	35389440	0	1
F12_S09	168	high	1.000000	35389440	0	1
F12_S10	168	high	1.000000	35389440	0	1
F12_S11	168	high	1.000000	35389440	0	1
F12_S12	168	high	1.000000	35389440	0	1
F12_S01	300	low	0.244444	9209088	87382	4
F12_S02	300	low	0.244444	9209088	87382	4
F12_S03	300	low	0.244444	9209088	87382	4
F12_S04	300	low	0.244444	9209088	87382	4
F12_S05	300	low	0.244444	9209088	87382	4
F12_S06	300	low	0.244444	9209088	87382	4
F12_S07	300	low	0.244444	9209088	87382	4
F12_S08	300	low	0.244444	9209088	87382	4
F12_S09	300	low	0.244444	9209088	87382	4
F12_S10	300	low	0.244444	9209088	87382	4
F12_S11	300	low	0.244444	9209088	87382	4
F12_S12	300	low	0.244444	9209088	87382	4
F12_S01	300	mid	0.500000	17729280	43691	2
F12_S02	300	mid	0.500000	17729280	43691	2
F12_S03	300	mid	0.500000	17729280	43691	2
F12_S04	300	mid	0.500000	17729280	43691	2
F12_S05	300	mid	0.500000	17729280	43691	2
F12_S06	300	mid	0.500000	17729280	43691	2
F12_S07	300	mid	0.500000	17729280	43691	2
F12_S08	300	mid	0.500000	17729280	43691	2
F12_S09	300	mid	0.500000	17729280	43691	2
F12_S10	300	mid	0.500000	17729280	43691	2
F12_S11	300	mid	0.500000	17729280	43691	2
F12_S12	300	mid	0.500000	17729280	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S01	300	high	1.000000	35389440	0	1
F12_S02	300	high	1.000000	35389440	0	1
F12_S03	300	high	1.000000	35389440	0	1
F12_S04	300	high	1.000000	35389440	0	1
F12_S05	300	high	1.000000	35389440	0	1
F12_S06	300	high	1.000000	35389440	0	1
F12_S07	300	high	1.000000	35389440	0	1
F12_S08	300	high	1.000000	35389440	0	1
F12_S09	300	high	1.000000	35389440	0	1
F12_S10	300	high	1.000000	35389440	0	1
F12_S11	300	high	1.000000	35389440	0	1
F12_S12	300	high	1.000000	35389440	0	1
F12_S01	1000	low	0.288889	10582272	87896	7
F12_S02	1000	low	0.288889	10582272	87896	7
F12_S03	1000	low	0.288889	10582272	87896	7
F12_S04	1000	low	0.288889	10582272	87896	7
F12_S05	1000	low	0.288889	10582272	87896	7
F12_S06	1000	low	0.288889	10582272	87896	7
F12_S07	1000	low	0.288889	10582272	87896	7
F12_S08	1000	low	0.288889	10582272	87896	7
F12_S09	1000	low	0.288889	10582272	87896	7
F12_S10	1000	low	0.288889	10582272	87896	7
F12_S11	1000	low	0.288889	10582272	87896	7
F12_S12	1000	low	0.288889	10582272	87896	7
F12_S01	1000	mid	0.500000	17729280	43691	2
F12_S02	1000	mid	0.500000	17729280	43691	2
F12_S03	1000	mid	0.500000	17729280	43691	2
F12_S04	1000	mid	0.500000	17729280	43691	2
F12_S05	1000	mid	0.500000	17729280	43691	2
F12_S06	1000	mid	0.500000	17729280	43691	2
F12_S07	1000	mid	0.500000	17729280	43691	2
F12_S08	1000	mid	0.500000	17729280	43691	2
F12_S09	1000	mid	0.500000	17729280	43691	2
F12_S10	1000	mid	0.500000	17729280	43691	2
F12_S11	1000	mid	0.500000	17729280	43691	2
F12_S12	1000	mid	0.500000	17729280	43691	2
F12_S01	1000	high	1.000000	35389440	0	1
F12_S02	1000	high	1.000000	35389440	0	1
F12_S03	1000	high	1.000000	35389440	0	1
F12_S04	1000	high	1.000000	35389440	0	1
F12_S05	1000	high	1.000000	35389440	0	1
F12_S06	1000	high	1.000000	35389440	0	1
F12_S07	1000	high	1.000000	35389440	0	1
F12_S08	1000	high	1.000000	35389440	0	1
F12_S09	1000	high	1.000000	35389440	0	1
F12_S10	1000	high	1.000000	35389440	0	1
F12_S11	1000	high	1.000000	35389440	0	1
F12_S12	1000	high	1.000000	35389440	0	1
F12_S01	3000	low	0.244444	9491712	87382	4
F12_S02	3000	low	0.244444	9491712	87382	4
F12_S03	3000	low	0.244444	9491712	87382	4
F12_S04	3000	low	0.244444	9491712	87382	4
F12_S05	3000	low	0.244444	9491712	87382	4
F12_S06	3000	low	0.244444	9491712	87382	4
F12_S07	3000	low	0.244444	9491712	87382	4
F12_S08	3000	low	0.244444	9491712	87382	4
F12_S09	3000	low	0.244444	9491712	87382	4
F12_S10	3000	low	0.244444	9491712	87382	4
F12_S11	3000	low	0.244444	9491712	87382	4
F12_S12	3000	low	0.244444	9491712	87382	4
F12_S01	3000	mid	0.500000	18247680	43691	2
F12_S02	3000	mid	0.500000	18247680	43691	2

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S03	3000	mid	0.500000	18247680	43691	2
F12_S04	3000	mid	0.500000	18247680	43691	2
F12_S05	3000	mid	0.500000	18247680	43691	2
F12_S06	3000	mid	0.500000	18247680	43691	2
F12_S07	3000	mid	0.500000	18247680	43691	2
F12_S08	3000	mid	0.500000	18247680	43691	2
F12_S09	3000	mid	0.500000	18247680	43691	2
F12_S10	3000	mid	0.500000	18247680	43691	2
F12_S11	3000	mid	0.500000	18247680	43691	2
F12_S12	3000	mid	0.500000	18247680	43691	2
F12_S01	3000	high	1.000000	35389440	0	1
F12_S02	3000	high	1.000000	35389440	0	1
F12_S03	3000	high	1.000000	35389440	0	1
F12_S04	3000	high	1.000000	35389440	0	1
F12_S05	3000	high	1.000000	35389440	0	1
F12_S06	3000	high	1.000000	35389440	0	1
F12_S07	3000	high	1.000000	35389440	0	1
F12_S08	3000	high	1.000000	35389440	0	1
F12_S09	3000	high	1.000000	35389440	0	1
F12_S10	3000	high	1.000000	35389440	0	1
F12_S11	3000	high	1.000000	35389440	0	1
F12_S12	3000	high	1.000000	35389440	0	1
F12_S01	10000	low	0.244444	9756672	87382	4
F12_S02	10000	low	0.244444	9756672	87382	4
F12_S03	10000	low	0.244444	9756672	87382	4
F12_S04	10000	low	0.244444	9756672	87382	4
F12_S05	10000	low	0.244444	9756672	87382	4
F12_S06	10000	low	0.244444	9756672	87382	4
F12_S07	10000	low	0.244444	9756672	87382	4
F12_S08	10000	low	0.244444	9756672	87382	4
F12_S09	10000	low	0.244444	9756672	87382	4
F12_S10	10000	low	0.244444	9756672	87382	4
F12_S11	10000	low	0.244444	9756672	87382	4
F12_S12	10000	low	0.244444	9756672	87382	4
F12_S01	10000	mid	0.500000	18247680	43691	2
F12_S02	10000	mid	0.500000	18247680	43691	2
F12_S03	10000	mid	0.500000	18247680	43691	2
F12_S04	10000	mid	0.500000	18247680	43691	2
F12_S05	10000	mid	0.500000	18247680	43691	2
F12_S06	10000	mid	0.500000	18247680	43691	2
F12_S07	10000	mid	0.500000	18247680	43691	2
F12_S08	10000	mid	0.500000	18247680	43691	2
F12_S09	10000	mid	0.500000	18247680	43691	2
F12_S10	10000	mid	0.500000	18247680	43691	2
F12_S11	10000	mid	0.500000	18247680	43691	2
F12_S12	10000	mid	0.500000	18247680	43691	2
F12_S01	10000	high	1.000000	35389440	0	1
F12_S02	10000	high	1.000000	35389440	0	1
F12_S03	10000	high	1.000000	35389440	0	1
F12_S04	10000	high	1.000000	35389440	0	1
F12_S05	10000	high	1.000000	35389440	0	1
F12_S06	10000	high	1.000000	35389440	0	1
F12_S07	10000	high	1.000000	35389440	0	1
F12_S08	10000	high	1.000000	35389440	0	1
F12_S09	10000	high	1.000000	35389440	0	1
F12_S10	10000	high	1.000000	35389440	0	1
F12_S11	10000	high	1.000000	35389440	0	1
F12_S12	10000	high	1.000000	35389440	0	1
F12_S01	30000	low	0.288889	11489280	99672	7
F12_S02	30000	low	0.288889	11489280	99672	7
F12_S03	30000	low	0.288889	11489280	99672	7
F12_S04	30000	low	0.288889	11489280	99672	7

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S05	30000	low	0.288889	11489280	99672	7
F12_S06	30000	low	0.288889	11489280	99672	7
F12_S07	30000	low	0.288889	11489280	99672	7
F12_S08	30000	low	0.288889	11489280	99672	7
F12_S09	30000	low	0.288889	11489280	99672	7
F12_S10	30000	low	0.288889	11489280	99672	7
F12_S11	30000	low	0.288889	11489280	99672	7
F12_S12	30000	low	0.288889	11489280	99672	7
F12_S01	30000	mid	0.500000	18800640	43691	2
F12_S02	30000	mid	0.500000	18800640	43691	2
F12_S03	30000	mid	0.500000	18800640	43691	2
F12_S04	30000	mid	0.500000	18800640	43691	2
F12_S05	30000	mid	0.500000	18800640	43691	2
F12_S06	30000	mid	0.500000	18800640	43691	2
F12_S07	30000	mid	0.500000	18800640	43691	2
F12_S08	30000	mid	0.500000	18800640	43691	2
F12_S09	30000	mid	0.500000	18800640	43691	2
F12_S10	30000	mid	0.500000	18800640	43691	2
F12_S11	30000	mid	0.500000	18800640	43691	2
F12_S12	30000	mid	0.500000	18800640	43691	2
F12_S01	30000	high	1.000000	35389440	0	1
F12_S02	30000	high	1.000000	35389440	0	1
F12_S03	30000	high	1.000000	35389440	0	1
F12_S04	30000	high	1.000000	35389440	0	1
F12_S05	30000	high	1.000000	35389440	0	1
F12_S06	30000	high	1.000000	35389440	0	1
F12_S07	30000	high	1.000000	35389440	0	1
F12_S08	30000	high	1.000000	35389440	0	1
F12_S09	30000	high	1.000000	35389440	0	1
F12_S10	30000	high	1.000000	35389440	0	1
F12_S11	30000	high	1.000000	35389440	0	1
F12_S12	30000	high	1.000000	35389440	0	1
F12_S01	100000	low	0.288889	11796480	142052	—
F12_S02	100000	low	0.288889	11796480	142052	—
F12_S03	100000	low	0.288889	11796480	142052	—
F12_S04	100000	low	0.288889	11796480	142052	—
F12_S05	100000	low	0.288889	11796480	142052	—
F12_S06	100000	low	0.288889	11796480	142052	—
F12_S07	100000	low	0.288889	11796480	142052	—
F12_S08	100000	low	0.288889	11796480	142052	—
F12_S09	100000	low	0.288889	11796480	142052	—
F12_S10	100000	low	0.288889	11796480	142052	—
F12_S11	100000	low	0.288889	11796480	142052	—
F12_S12	100000	low	0.288889	11796480	142052	—
F12_S01	100000	mid	0.622222	22855680	95575	8
F12_S02	100000	mid	0.622222	22855680	95575	8
F12_S03	100000	mid	0.622222	22855680	95575	8
F12_S04	100000	mid	0.622222	22855680	95575	8
F12_S05	100000	mid	0.622222	22855680	95575	8
F12_S06	100000	mid	0.622222	22855680	95575	8
F12_S07	100000	mid	0.622222	22855680	95575	8
F12_S08	100000	mid	0.622222	22855680	95575	8
F12_S09	100000	mid	0.622222	22855680	95575	8
F12_S10	100000	mid	0.622222	22855680	95575	8
F12_S11	100000	mid	0.622222	22855680	95575	8
F12_S12	100000	mid	0.622222	22855680	95575	8
F12_S01	100000	high	1.000000	35389440	0	1
F12_S02	100000	high	1.000000	35389440	0	1
F12_S03	100000	high	1.000000	35389440	0	1
F12_S04	100000	high	1.000000	35389440	0	1
F12_S05	100000	high	1.000000	35389440	0	1
F12_S06	100000	high	1.000000	35389440	0	1

Session	v	Cap.	Core frac.	Payload (B)	Max Q (B)	Period
F12_S07	100000	high	1.000000	35389440	0	1
F12_S08	100000	high	1.000000	35389440	0	1
F12_S09	100000	high	1.000000	35389440	0	1
F12_S10	100000	high	1.000000	35389440	0	1
F12_S11	100000	high	1.000000	35389440	0	1
F12_S12	100000	high	1.000000	35389440	0	1

VI. COMPLETE PHASE-RANDOMIZATION SENSITIVITY

Probe phases are independently assigned from fixed seeds. Probes exchange neither action nor queue state, and no joint admission is introduced.

TABLE IV: Complete desynchronization software-sensitivity results (240 retained rows).

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090901	low	desynchronized_independent	0.333333	0.000000	131029	595/193/691
2026090901	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090901	low	central_coordinated	0.000000	–	16341	0/0/0
2026090901	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090901	mid	desynchronized_independent	0.666667	0.505556	130985	595/193/691
2026090901	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090901	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090901	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090901	high	desynchronized_independent	1.000000	1.000000	121111	595/193/691
2026090901	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090901	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090901	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090902	low	desynchronized_independent	0.333333	0.000000	131029	403/632/306
2026090902	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090902	low	central_coordinated	0.000000	–	16341	0/0/0
2026090902	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090902	mid	desynchronized_independent	0.666667	0.500000	130985	403/632/306
2026090902	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090902	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090902	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090902	high	desynchronized_independent	1.000000	1.000000	130876	403/632/306
2026090902	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090902	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090902	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090903	low	desynchronized_independent	0.333333	0.000000	131029	489/690/996
2026090903	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090903	low	central_coordinated	0.000000	–	16341	0/0/0
2026090903	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090903	mid	desynchronized_independent	0.666667	0.500000	130985	489/690/996
2026090903	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090903	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090903	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090903	high	desynchronized_independent	1.000000	1.000000	1180	489/690/996
2026090903	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090903	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090903	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090904	low	desynchronized_independent	0.333333	0.000000	131028	945/680/799
2026090904	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090904	low	central_coordinated	0.000000	–	16341	0/0/0
2026090904	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090904	mid	desynchronized_independent	0.666667	0.500000	130985	945/680/799
2026090904	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090904	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090904	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090904	high	desynchronized_independent	1.000000	1.000000	21234	945/680/799
2026090904	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090904	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090904	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090905	low	desynchronized_independent	0.333333	0.000000	131028	154/172/828
2026090905	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090905	low	central_coordinated	0.000000	–	16341	0/0/0
2026090905	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090905	mid	desynchronized_independent	0.666667	0.505556	130985	154/172/828
2026090905	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090905	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090905	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090905	high	desynchronized_independent	1.000000	1.000000	130876	154/172/828
2026090905	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090905	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090905	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090906	low	desynchronized_independent	0.333333	0.000000	131029	60/626/317
2026090906	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090906	low	central_coordinated	0.000000	–	16341	0/0/0
2026090906	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090906	mid	desynchronized_independent	0.666667	0.505556	130985	60/626/317
2026090906	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090906	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090906	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090906	high	desynchronized_independent	1.000000	1.000000	130876	60/626/317
2026090906	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090906	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090906	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090907	low	desynchronized_independent	0.333333	0.000000	131029	753/955/576
2026090907	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090907	low	central_coordinated	0.000000	–	16341	0/0/0
2026090907	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090907	mid	desynchronized_independent	0.666667	0.500000	130985	753/955/576
2026090907	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090907	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090907	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090907	high	desynchronized_independent	1.000000	1.000000	17302	753/955/576
2026090907	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090907	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090907	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090908	low	desynchronized_independent	0.333333	0.000000	131029	884/483/389
2026090908	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090908	low	central_coordinated	0.000000	–	16341	0/0/0
2026090908	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090908	mid	desynchronized_independent	0.666667	0.500000	130985	884/483/389
2026090908	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090908	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090908	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090908	high	desynchronized_independent	1.000000	1.000000	71828	884/483/389
2026090908	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090908	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090908	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090909	low	desynchronized_independent	0.333333	0.000000	131029	530/348/834
2026090909	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090909	low	central_coordinated	0.000000	–	16341	0/0/0
2026090909	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090909	mid	desynchronized_independent	0.666667	0.500000	130985	530/348/834
2026090909	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090909	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090909	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090909	high	desynchronized_independent	1.000000	1.000000	64881	530/348/834
2026090909	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090909	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090909	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090910	low	desynchronized_independent	0.333333	0.000000	131028	611/131/96
2026090910	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090910	low	central_coordinated	0.000000	–	16341	0/0/0
2026090910	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090910	mid	desynchronized_independent	0.666667	0.505556	130985	611/131/96
2026090910	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090910	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090910	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090910	high	desynchronized_independent	1.000000	1.000000	130876	611/131/96
2026090910	high	synchronized_independent	1.000000	1.000000	130941	0/0/0

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090910	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090910	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090911	low	desynchronized_independent	0.333333	0.000000	131029	286/697/613
2026090911	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090911	low	central_coordinated	0.000000	–	16341	0/0/0
2026090911	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090911	mid	desynchronized_independent	0.666667	0.505556	130985	286/697/613
2026090911	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090911	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090911	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090911	high	desynchronized_independent	1.000000	1.000000	118752	286/697/613
2026090911	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090911	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090911	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090912	low	desynchronized_independent	0.333333	0.000000	131028	72/898/820
2026090912	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090912	low	central_coordinated	0.000000	–	16341	0/0/0
2026090912	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090912	mid	desynchronized_independent	0.666667	0.505556	130985	72/898/820
2026090912	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090912	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090912	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090912	high	desynchronized_independent	1.000000	1.000000	102368	72/898/820
2026090912	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090912	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090912	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090913	low	desynchronized_independent	0.333333	0.000000	131028	31/997/848
2026090913	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090913	low	central_coordinated	0.000000	–	16341	0/0/0
2026090913	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090913	mid	desynchronized_independent	0.666667	0.505556	130985	31/997/848
2026090913	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090913	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090913	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090913	high	desynchronized_independent	1.000000	1.000000	118490	31/997/848
2026090913	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090913	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090913	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090914	low	desynchronized_independent	0.333333	0.000000	131029	379/396/678
2026090914	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090914	low	central_coordinated	0.000000	–	16341	0/0/0
2026090914	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090914	mid	desynchronized_independent	0.666667	0.500000	130985	379/396/678
2026090914	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090914	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090914	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090914	high	desynchronized_independent	1.000000	1.000000	126223	379/396/678
2026090914	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090914	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090914	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090915	low	desynchronized_independent	0.333333	0.000000	131028	212/604/213
2026090915	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090915	low	central_coordinated	0.000000	–	16341	0/0/0
2026090915	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090915	mid	desynchronized_independent	0.666667	0.505556	130985	212/604/213
2026090915	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090915	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090915	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090915	high	desynchronized_independent	1.000000	1.000000	130876	212/604/213
2026090915	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090915	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090915	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0

Seed	Cap.	Method	Core frac.	On-time	Max Q (B)	Phases (ms)
2026090916	low	desynchronized_independent	0.333333	0.000000	131029	586/124/90
2026090916	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090916	low	central_coordinated	0.000000	–	16341	0/0/0
2026090916	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090916	mid	desynchronized_independent	0.666667	0.505556	130985	586/124/90
2026090916	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090916	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090916	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090916	high	desynchronized_independent	1.000000	1.000000	130876	586/124/90
2026090916	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090916	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090916	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090917	low	desynchronized_independent	0.333333	0.000000	131029	13/5/654
2026090917	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090917	low	central_coordinated	0.000000	–	16341	0/0/0
2026090917	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090917	mid	desynchronized_independent	0.666667	0.500000	130985	13/5/654
2026090917	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090917	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090917	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090917	high	desynchronized_independent	1.000000	1.000000	130876	13/5/654
2026090917	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090917	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090917	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090918	low	desynchronized_independent	0.333333	0.000000	131029	303/966/432
2026090918	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090918	low	central_coordinated	0.000000	–	16341	0/0/0
2026090918	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090918	mid	desynchronized_independent	0.666667	0.500000	130985	303/966/432
2026090918	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090918	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090918	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090918	high	desynchronized_independent	1.000000	1.000000	91882	303/966/432
2026090918	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090918	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090918	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090919	low	desynchronized_independent	0.333333	0.000000	131029	94/427/937
2026090919	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090919	low	central_coordinated	0.000000	–	16341	0/0/0
2026090919	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090919	mid	desynchronized_independent	0.666667	0.500000	130985	94/427/937
2026090919	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090919	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090919	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090919	high	desynchronized_independent	1.000000	1.000000	93848	94/427/937
2026090919	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090919	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090919	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0
2026090920	low	desynchronized_independent	0.333333	0.000000	131029	699/148/926
2026090920	low	synchronized_independent	0.244444	0.000000	131029	0/0/0
2026090920	low	central_coordinated	0.000000	–	16341	0/0/0
2026090920	low	shielded_coordinated	0.000000	–	16341	0/0/0
2026090920	mid	desynchronized_independent	0.666667	0.505556	130985	699/148/926
2026090920	mid	synchronized_independent	0.500000	0.000000	130985	0/0/0
2026090920	mid	central_coordinated	0.333333	1.000000	130985	0/0/0
2026090920	mid	shielded_coordinated	0.333333	1.000000	130985	0/0/0
2026090920	high	desynchronized_independent	1.000000	1.000000	72483	699/148/926
2026090920	high	synchronized_independent	1.000000	1.000000	130941	0/0/0
2026090920	high	central_coordinated	1.000000	1.000000	130941	0/0/0
2026090920	high	shielded_coordinated	1.000000	1.000000	130941	0/0/0

VII. COMPLETE SCHEDULE-ANCHOR RECONSTRUCTION

Post-release latency is result return minus actual event creation. Nominal latency is result return minus scheduled release. The 1-s quantity in the main paper is a post-release service SLO, not a complete schedule-to-result deadline.

TABLE V: Complete accepted-hardware timing reconstruction (648 retained cells; latency in ms).

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S01	fixed_local	low	P1	0	–	–	1862.2	1920.9	1924.1
F12_S01	fixed_local	low	P2	0	–	–	760.3	831.8	834.1
F12_S01	fixed_local	low	P3	0	–	–	86.3	155.7	162.0
F12_S01	fixed_local	mid	P1	0	–	–	1828.4	1979.1	1980.6
F12_S01	fixed_local	mid	P2	0	–	–	636.8	775.7	780.4
F12_S01	fixed_local	mid	P3	0	–	–	87.5	165.1	172.5
F12_S01	fixed_local	high	P1	0	–	–	1827.5	1863.0	1913.4
F12_S01	fixed_local	high	P2	0	–	–	637.0	671.0	673.3
F12_S01	fixed_local	high	P3	0	–	–	84.4	156.5	163.5
F12_S01	fixed_summary	low	P1	0	–	–	1952.6	2002.0	2006.5
F12_S01	fixed_summary	low	P2	0	–	–	712.1	750.1	753.4
F12_S01	fixed_summary	low	P3	0	–	–	84.0	156.4	163.8
F12_S01	fixed_summary	mid	P1	0	–	–	1922.6	1997.1	2001.5
F12_S01	fixed_summary	mid	P2	0	–	–	716.0	804.2	806.5
F12_S01	fixed_summary	mid	P3	0	–	–	85.2	153.6	160.0
F12_S01	fixed_summary	high	P1	0	–	–	1925.6	1978.0	1981.4
F12_S01	fixed_summary	high	P2	0	–	–	745.5	802.9	806.8
F12_S01	fixed_summary	high	P3	0	–	–	88.3	159.6	166.3
F12_S01	fixed_full	low	P1	90	0.0000	0.0000	1887.7	1981.0	1983.3
F12_S01	fixed_full	low	P2	90	0.0000	0.0000	736.5	827.9	830.4
F12_S01	fixed_full	low	P3	90	0.0000	0.0000	79.3	139.5	145.3
F12_S01	fixed_full	mid	P1	90	0.0000	0.0000	1859.1	2483.1	2486.7
F12_S01	fixed_full	mid	P2	90	0.0000	0.0000	686.7	1341.8	1346.6
F12_S01	fixed_full	mid	P3	90	0.0000	0.0000	79.2	141.9	148.5
F12_S01	fixed_full	high	P1	90	0.9667	0.0000	2024.3	2101.8	2105.2
F12_S01	fixed_full	high	P2	90	0.9222	0.0000	830.5	948.8	952.3
F12_S01	fixed_full	high	P3	90	1.0000	1.0000	78.6	141.2	145.6
F12_S01	independent_dpp	low	P1	22	0.0000	0.0000	1851.5	2183.8	2301.2
F12_S01	independent_dpp	low	P2	22	0.0000	0.0000	661.8	792.4	1053.9
F12_S01	independent_dpp	low	P3	22	0.0000	0.0000	82.3	152.1	156.3
F12_S01	independent_dpp	mid	P1	45	0.0000	0.0000	1998.0	2108.8	2111.0
F12_S01	independent_dpp	mid	P2	45	0.0000	0.0000	754.9	822.0	824.8
F12_S01	independent_dpp	mid	P3	45	0.0000	0.0000	81.4	146.9	152.4
F12_S01	independent_dpp	high	P1	90	0.9444	0.0000	1967.9	2006.6	2009.6
F12_S01	independent_dpp	high	P2	90	0.9556	0.0000	797.6	836.1	841.8
F12_S01	independent_dpp	high	P3	90	1.0000	1.0000	78.4	136.1	140.8
F12_S01	central_dpp	low	P1	0	–	–	2049.7	2151.4	2153.5
F12_S01	central_dpp	low	P2	0	–	–	858.4	942.6	947.1
F12_S01	central_dpp	low	P3	0	–	–	83.2	148.0	154.0
F12_S01	central_dpp	mid	P1	30	0.8000	0.0000	1910.4	1992.9	1995.4
F12_S01	central_dpp	mid	P2	30	0.8000	0.0000	732.4	809.6	813.0
F12_S01	central_dpp	mid	P3	30	1.0000	1.0000	83.1	152.5	159.3
F12_S01	central_dpp	high	P1	90	1.0000	0.0000	2070.5	2105.4	2109.1
F12_S01	central_dpp	high	P2	90	0.9889	0.0000	907.5	949.5	953.2
F12_S01	central_dpp	high	P3	90	1.0000	1.0000	78.5	144.9	150.8
F12_S01	shielded_dpp	low	P1	0	–	–	1771.7	1832.4	1835.5
F12_S01	shielded_dpp	low	P2	0	–	–	678.1	749.6	786.5
F12_S01	shielded_dpp	low	P3	0	–	–	87.8	156.7	162.3
F12_S01	shielded_dpp	mid	P1	30	0.9000	0.0000	1934.5	2160.6	2163.7
F12_S01	shielded_dpp	mid	P2	30	1.0000	0.0000	856.4	964.9	967.9
F12_S01	shielded_dpp	mid	P3	30	1.0000	1.0000	86.3	149.8	155.9
F12_S01	shielded_dpp	high	P1	90	0.9333	0.0000	1979.2	2065.8	2070.3
F12_S01	shielded_dpp	high	P2	90	0.9444	0.0000	876.4	913.1	974.2
F12_S01	shielded_dpp	high	P3	90	1.0000	1.0000	79.7	136.0	140.8
F12_S02	fixed_local	low	P1	0	–	–	1886.4	2021.6	2024.1
F12_S02	fixed_local	low	P2	0	–	–	594.0	740.2	743.2
F12_S02	fixed_local	low	P3	0	–	–	88.2	164.6	171.5
F12_S02	fixed_local	mid	P1	0	–	–	2003.6	2046.7	2051.4
F12_S02	fixed_local	mid	P2	0	–	–	717.1	761.0	764.1
F12_S02	fixed_local	mid	P3	0	–	–	92.2	161.7	168.4
F12_S02	fixed_local	high	P1	0	–	–	1896.5	1938.2	1941.3
F12_S02	fixed_local	high	P2	0	–	–	737.1	775.7	778.9
F12_S02	fixed_local	high	P3	0	–	–	91.3	160.5	167.6
F12_S02	fixed_summary	low	P1	0	–	–	1725.1	1764.9	1767.8
F12_S02	fixed_summary	low	P2	0	–	–	550.6	585.7	589.6
F12_S02	fixed_summary	low	P3	0	–	–	90.6	160.8	167.3
F12_S02	fixed_summary	mid	P1	0	–	–	1915.3	2065.5	2072.1
F12_S02	fixed_summary	mid	P2	0	–	–	677.3	1103.5	1106.6
F12_S02	fixed_summary	mid	P3	0	–	–	85.8	156.6	162.7
F12_S02	fixed_summary	high	P1	0	–	–	1878.3	1949.3	1959.8
F12_S02	fixed_summary	high	P2	0	–	–	713.1	787.0	798.8
F12_S02	fixed_summary	high	P3	0	–	–	89.0	158.6	165.8
F12_S02	fixed_full	low	P1	90	0.0000	0.0000	1929.0	2110.4	2114.6
F12_S02	fixed_full	low	P2	90	0.0000	0.0000	763.0	876.5	879.5

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S02	fixed_full	low	P3	90	0.0000	0.0000	78.2	135.9	142.5
F12_S02	fixed_full	mid	P1	90	0.0000	0.0000	2251.6	2305.8	2309.9
F12_S02	fixed_full	mid	P2	90	0.0000	0.0000	616.5	733.8	750.5
F12_S02	fixed_full	mid	P3	90	0.0000	0.0000	76.4	136.6	142.5
F12_S02	fixed_full	high	P1	90	1.0000	0.0000	1969.5	1996.6	2001.5
F12_S02	fixed_full	high	P2	90	0.9889	0.0000	827.2	865.1	867.9
F12_S02	fixed_full	high	P3	90	1.0000	1.0000	80.0	145.5	151.7
F12_S02	independent_dpp	low	P1	22	0.0000	0.0000	1906.4	2060.6	2064.0
F12_S02	independent_dpp	low	P2	22	0.0000	0.0000	748.0	899.5	904.3
F12_S02	independent_dpp	low	P3	22	0.0000	0.0000	78.9	142.0	147.5
F12_S02	independent_dpp	mid	P1	45	0.0000	0.0000	1980.3	2013.9	2015.6
F12_S02	independent_dpp	mid	P2	45	0.0000	0.0000	703.8	732.1	734.0
F12_S02	independent_dpp	mid	P3	45	0.0000	0.0000	79.5	147.8	152.5
F12_S02	independent_dpp	high	P1	90	0.9111	0.0000	1982.0	2052.8	2056.0
F12_S02	independent_dpp	high	P2	90	0.9333	0.0000	787.2	831.0	832.9
F12_S02	independent_dpp	high	P3	90	1.0000	1.0000	78.0	137.5	141.7
F12_S02	central_dpp	low	P1	0	–	–	2017.6	2154.2	2157.5
F12_S02	central_dpp	low	P2	0	–	–	738.5	861.7	865.2
F12_S02	central_dpp	low	P3	0	–	–	87.9	158.3	164.1
F12_S02	central_dpp	mid	P1	30	0.7333	0.0000	1936.7	2071.6	2074.8
F12_S02	central_dpp	mid	P2	30	0.9333	0.0000	881.1	913.6	916.2
F12_S02	central_dpp	mid	P3	30	1.0000	1.0000	86.2	150.4	156.4
F12_S02	central_dpp	high	P1	90	0.9222	0.0000	1942.0	1995.4	2000.0
F12_S02	central_dpp	high	P2	90	0.9444	0.0000	816.7	862.3	865.6
F12_S02	central_dpp	high	P3	90	1.0000	1.0000	81.3	145.8	151.8
F12_S02	shielded_dpp	low	P1	0	–	–	1931.4	1995.7	1999.1
F12_S02	shielded_dpp	low	P2	0	–	–	842.4	903.9	908.0
F12_S02	shielded_dpp	low	P3	0	–	–	86.6	160.6	168.3
F12_S02	shielded_dpp	mid	P1	30	0.9000	0.0000	2120.6	2150.6	2151.7
F12_S02	shielded_dpp	mid	P2	30	0.9000	0.0000	793.9	826.4	828.9
F12_S02	shielded_dpp	mid	P3	30	1.0000	1.0000	80.3	148.0	155.9
F12_S02	shielded_dpp	high	P1	90	0.9444	0.0000	1954.9	2007.4	2039.4
F12_S02	shielded_dpp	high	P2	90	0.9333	0.0000	781.0	858.6	873.6
F12_S02	shielded_dpp	high	P3	90	1.0000	1.0000	81.5	140.4	145.8
F12_S03	fixed_local	low	P1	0	–	–	1892.3	1942.8	1950.8
F12_S03	fixed_local	low	P2	0	–	–	676.8	732.8	744.0
F12_S03	fixed_local	low	P3	0	–	–	90.0	162.6	169.3
F12_S03	fixed_local	mid	P1	0	–	–	1823.3	1868.5	1871.1
F12_S03	fixed_local	mid	P2	0	–	–	736.6	837.7	840.2
F12_S03	fixed_local	mid	P3	0	–	–	89.2	160.3	166.4
F12_S03	fixed_local	high	P1	0	–	–	1939.5	2052.7	2061.3
F12_S03	fixed_local	high	P2	0	–	–	724.4	838.4	852.7
F12_S03	fixed_local	high	P3	0	–	–	90.8	167.3	174.3
F12_S03	fixed_summary	low	P1	0	–	–	1926.4	1967.6	1971.1
F12_S03	fixed_summary	low	P2	0	–	–	710.5	757.2	762.6
F12_S03	fixed_summary	low	P3	0	–	–	90.1	166.0	172.4
F12_S03	fixed_summary	mid	P1	0	–	–	2190.5	2234.0	2237.6
F12_S03	fixed_summary	mid	P2	0	–	–	877.1	918.5	922.6
F12_S03	fixed_summary	mid	P3	0	–	–	90.8	162.8	169.3
F12_S03	fixed_summary	high	P1	0	–	–	2024.1	2062.5	2066.0
F12_S03	fixed_summary	high	P2	0	–	–	748.8	787.7	790.7
F12_S03	fixed_summary	high	P3	0	–	–	89.3	160.2	166.3
F12_S03	fixed_full	low	P1	90	0.0000	0.0000	1963.8	2094.5	2097.1
F12_S03	fixed_full	low	P2	90	0.0000	0.0000	864.3	926.3	947.9
F12_S03	fixed_full	low	P3	90	0.0000	0.0000	78.2	147.4	153.9
F12_S03	fixed_full	mid	P1	90	0.0000	0.0000	1903.6	1975.4	1976.5
F12_S03	fixed_full	mid	P2	90	0.0000	0.0000	752.0	824.8	829.8
F12_S03	fixed_full	mid	P3	90	0.0000	0.0000	80.7	141.3	147.6
F12_S03	fixed_full	high	P1	90	0.9667	0.0000	2006.1	2050.0	2098.4
F12_S03	fixed_full	high	P2	90	0.9333	0.0000	847.8	892.2	1004.3
F12_S03	fixed_full	high	P3	90	1.0000	1.0000	86.9	150.9	156.0
F12_S03	independent_dpp	low	P1	22	0.0000	0.0000	1937.5	2023.7	2025.5
F12_S03	independent_dpp	low	P2	22	0.0000	0.0000	755.3	832.0	841.6
F12_S03	independent_dpp	low	P3	22	0.0000	0.0000	82.0	151.6	158.6
F12_S03	independent_dpp	mid	P1	45	0.0000	0.0000	7335.7	7390.7	7395.4
F12_S03	independent_dpp	mid	P2	45	0.0000	0.0000	6236.0	6290.3	6295.7
F12_S03	independent_dpp	mid	P3	45	0.0000	0.0000	4851.8	4918.9	4924.2
F12_S03	independent_dpp	high	P1	90	0.9889	0.0000	1867.0	2014.7	2018.8
F12_S03	independent_dpp	high	P2	90	0.9556	0.0000	699.5	814.5	817.0
F12_S03	independent_dpp	high	P3	90	1.0000	1.0000	74.3	134.1	139.7
F12_S03	central_dpp	low	P1	0	–	–	1984.5	2024.7	2030.2
F12_S03	central_dpp	low	P2	0	–	–	838.4	875.1	878.3
F12_S03	central_dpp	low	P3	0	–	–	90.1	161.4	168.3
F12_S03	central_dpp	mid	P1	30	0.8667	0.0000	1949.1	2028.9	2031.1
F12_S03	central_dpp	mid	P2	30	0.7667	0.0000	915.3	954.2	957.1
F12_S03	central_dpp	mid	P3	30	1.0000	1.0000	81.0	147.8	154.9
F12_S03	central_dpp	high	P1	90	0.9667	0.0000	1963.1	2023.1	2025.9
F12_S03	central_dpp	high	P2	90	1.0000	0.0000	856.1	892.5	896.6
F12_S03	central_dpp	high	P3	90	1.0000	1.0000	73.2	133.8	140.1
F12_S03	shielded_dpp	low	P1	0	–	–	2074.6	2116.7	2120.3
F12_S03	shielded_dpp	low	P2	0	–	–	627.9	665.5	668.1
F12_S03	shielded_dpp	low	P3	0	–	–	84.1	154.8	160.5
F12_S03	shielded_dpp	mid	P1	30	0.8000	0.0000	2003.3	2147.9	2150.1

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S03	shielded_dpp	mid	P2	30	0.8333	0.0000	758.1	862.1	892.8
F12_S03	shielded_dpp	mid	P3	30	1.0000	1.0000	86.6	151.2	156.3
F12_S03	shielded_dpp	high	P1	90	0.9333	0.0000	2057.5	2130.4	2139.2
F12_S03	shielded_dpp	high	P2	90	0.9778	0.0000	939.6	989.1	992.6
F12_S03	shielded_dpp	high	P3	90	1.0000	1.0000	77.9	142.0	146.9
F12_S04	fixed_local	low	P1	0	–	–	1879.6	1975.4	1977.9
F12_S04	fixed_local	low	P2	0	–	–	657.9	744.4	746.8
F12_S04	fixed_local	low	P3	0	–	–	91.4	163.7	170.4
F12_S04	fixed_local	mid	P1	0	–	–	1944.5	2017.8	2022.3
F12_S04	fixed_local	mid	P2	0	–	–	724.9	795.4	798.7
F12_S04	fixed_local	mid	P3	0	–	–	95.3	169.4	176.3
F12_S04	fixed_local	high	P1	0	–	–	1896.6	1934.6	1938.1
F12_S04	fixed_local	high	P2	0	–	–	743.8	780.2	783.8
F12_S04	fixed_local	high	P3	0	–	–	87.9	160.4	166.0
F12_S04	fixed_summary	low	P1	0	–	–	1958.8	2049.4	2059.2
F12_S04	fixed_summary	low	P2	0	–	–	719.0	800.6	806.0
F12_S04	fixed_summary	low	P3	0	–	–	87.6	156.2	162.2
F12_S04	fixed_summary	mid	P1	0	–	–	1929.8	2022.4	2027.0
F12_S04	fixed_summary	mid	P2	0	–	–	706.0	830.5	834.7
F12_S04	fixed_summary	mid	P3	0	–	–	89.1	158.3	164.2
F12_S04	fixed_summary	high	P1	0	–	–	1973.7	2015.9	2019.2
F12_S04	fixed_summary	high	P2	0	–	–	595.9	629.4	630.5
F12_S04	fixed_summary	high	P3	0	–	–	88.2	155.6	161.3
F12_S04	fixed_full	low	P1	90	0.0000	0.0000	2135.3	2299.8	2301.8
F12_S04	fixed_full	low	P2	90	0.0000	0.0000	843.4	1033.6	1036.2
F12_S04	fixed_full	low	P3	90	0.0000	0.0000	78.4	140.7	146.6
F12_S04	fixed_full	mid	P1	90	0.0000	0.0000	1990.7	2057.3	2060.3
F12_S04	fixed_full	mid	P2	90	0.0000	0.0000	864.7	897.3	900.8
F12_S04	fixed_full	mid	P3	90	0.0000	0.0000	78.5	141.6	146.8
F12_S04	fixed_full	high	P1	90	0.9556	0.0000	1968.2	2168.1	2172.8
F12_S04	fixed_full	high	P2	90	0.9222	0.0000	746.4	946.3	950.5
F12_S04	fixed_full	high	P3	90	1.0000	1.0000	78.6	142.2	146.8
F12_S04	independent_dpp	low	P1	22	0.0000	0.0000	2113.4	2196.6	2199.2
F12_S04	independent_dpp	low	P2	22	0.0000	0.0000	1004.2	1036.6	1038.8
F12_S04	independent_dpp	low	P3	22	0.0000	0.0000	85.1	154.4	159.8
F12_S04	independent_dpp	mid	P1	45	0.0000	0.0000	1990.7	2228.1	2232.6
F12_S04	independent_dpp	mid	P2	45	0.0000	0.0000	837.2	975.7	977.9
F12_S04	independent_dpp	mid	P3	45	0.0000	0.0000	77.9	143.7	149.7
F12_S04	independent_dpp	high	P1	90	0.9444	0.0000	2007.0	2091.5	2094.1
F12_S04	independent_dpp	high	P2	90	0.9556	0.0000	806.0	927.9	929.9
F12_S04	independent_dpp	high	P3	90	1.0000	1.0000	80.4	145.8	151.8
F12_S04	central_dpp	low	P1	0	–	–	1985.7	2026.6	2029.4
F12_S04	central_dpp	low	P2	0	–	–	669.0	706.4	709.7
F12_S04	central_dpp	low	P3	0	–	–	85.8	158.2	163.9
F12_S04	central_dpp	mid	P1	30	0.6667	0.0000	1908.9	1971.8	1982.7
F12_S04	central_dpp	mid	P2	30	0.7667	0.0000	747.3	842.3	846.1
F12_S04	central_dpp	mid	P3	30	1.0000	1.0000	86.9	156.1	162.2
F12_S04	central_dpp	high	P1	90	0.9556	0.0000	2089.1	2217.9	2220.9
F12_S04	central_dpp	high	P2	90	0.9889	0.0000	855.7	922.1	923.9
F12_S04	central_dpp	high	P3	90	1.0000	1.0000	74.0	133.8	139.0
F12_S04	shielded_dpp	low	P1	0	–	–	1873.0	1929.6	1934.2
F12_S04	shielded_dpp	low	P2	0	–	–	648.1	713.7	716.7
F12_S04	shielded_dpp	low	P3	0	–	–	85.0	156.3	162.7
F12_S04	shielded_dpp	mid	P1	30	0.7667	0.0000	1978.4	2044.0	2047.9
F12_S04	shielded_dpp	mid	P2	30	0.7667	0.0000	795.4	884.6	887.1
F12_S04	shielded_dpp	mid	P3	30	1.0000	1.0000	81.0	144.6	150.5
F12_S04	shielded_dpp	high	P1	90	0.9778	0.0000	1987.3	2025.7	2029.0
F12_S04	shielded_dpp	high	P2	90	0.9111	0.0000	724.4	854.0	858.5
F12_S04	shielded_dpp	high	P3	90	1.0000	1.0000	78.4	144.4	150.5
F12_S05	fixed_local	low	P1	0	–	–	2011.8	2048.3	2050.5
F12_S05	fixed_local	low	P2	0	–	–	735.0	769.7	771.7
F12_S05	fixed_local	low	P3	0	–	–	88.0	158.3	164.5
F12_S05	fixed_local	mid	P1	0	–	–	1826.1	1896.2	1901.4
F12_S05	fixed_local	mid	P2	0	–	–	665.2	704.7	706.8
F12_S05	fixed_local	mid	P3	0	–	–	86.6	158.2	164.3
F12_S05	fixed_local	high	P1	0	–	–	1917.4	1963.7	1966.4
F12_S05	fixed_local	high	P2	0	–	–	763.8	800.2	802.8
F12_S05	fixed_local	high	P3	0	–	–	90.4	169.1	175.5
F12_S05	fixed_summary	low	P1	0	–	–	1836.5	1895.6	1899.3
F12_S05	fixed_summary	low	P2	0	–	–	624.0	695.4	698.3
F12_S05	fixed_summary	low	P3	0	–	–	88.9	160.6	166.5
F12_S05	fixed_summary	mid	P1	0	–	–	1787.3	1904.6	1908.2
F12_S05	fixed_summary	mid	P2	0	–	–	749.4	864.3	868.8
F12_S05	fixed_summary	mid	P3	0	–	–	91.0	163.6	170.1
F12_S05	fixed_summary	high	P1	0	–	–	1869.3	1931.6	1935.0
F12_S05	fixed_summary	high	P2	0	–	–	712.6	774.4	779.4
F12_S05	fixed_summary	high	P3	0	–	–	86.0	152.7	159.9
F12_S05	fixed_full	low	P1	90	0.0000	0.0000	2017.7	2049.2	2051.7
F12_S05	fixed_full	low	P2	90	0.0000	0.0000	779.2	840.4	844.0
F12_S05	fixed_full	low	P3	90	0.0000	0.0000	76.6	140.7	146.2
F12_S05	fixed_full	mid	P1	90	0.0000	0.0000	1920.6	2001.0	2049.4
F12_S05	fixed_full	mid	P2	90	0.0000	0.0000	746.0	797.5	843.3
F12_S05	fixed_full	mid	P3	90	0.0000	0.0000	78.4	144.6	151.0

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S05	fixed_full	high	P1	90	0.9111	0.0000	1861.7	1922.9	1926.9
F12_S05	fixed_full	high	P2	90	0.9222	0.0000	777.4	829.5	831.5
F12_S05	fixed_full	high	P3	90	1.0000	1.0000	81.1	140.9	146.8
F12_S05	independent_dpp	low	P1	22	0.0000	0.0000	1901.3	1976.4	1981.7
F12_S05	independent_dpp	low	P2	22	0.0000	0.0000	654.1	712.2	715.8
F12_S05	independent_dpp	low	P3	22	0.0000	0.0000	81.8	145.1	151.4
F12_S05	independent_dpp	mid	P1	45	0.0000	0.0000	1841.5	1921.5	1925.4
F12_S05	independent_dpp	mid	P2	45	0.0000	0.0000	675.7	737.0	747.7
F12_S05	independent_dpp	mid	P3	45	0.0000	0.0000	74.9	137.5	142.5
F12_S05	independent_dpp	high	P1	90	0.9778	0.0000	1962.0	2083.5	2087.9
F12_S05	independent_dpp	high	P2	90	0.9778	0.0000	886.6	920.4	923.2
F12_S05	independent_dpp	high	P3	90	1.0000	1.0000	76.8	139.6	145.5
F12_S05	central_dpp	low	P1	0	–	–	1957.7	1999.1	2002.0
F12_S05	central_dpp	low	P2	0	–	–	769.8	811.2	814.7
F12_S05	central_dpp	low	P3	0	–	–	85.3	153.6	160.1
F12_S05	central_dpp	mid	P1	30	1.0000	0.0000	1912.7	1940.1	1942.0
F12_S05	central_dpp	mid	P2	30	0.8000	0.0000	793.2	835.2	837.4
F12_S05	central_dpp	mid	P3	30	1.0000	1.0000	82.5	150.1	155.5
F12_S05	central_dpp	high	P1	90	0.9111	0.0000	1946.2	1994.5	1999.5
F12_S05	central_dpp	high	P2	90	0.9778	0.0000	792.1	861.7	865.5
F12_S05	central_dpp	high	P3	90	1.0000	1.0000	77.4	141.8	146.7
F12_S05	shielded_dpp	low	P1	0	–	–	1893.0	1978.6	1982.3
F12_S05	shielded_dpp	low	P2	0	–	–	673.0	754.2	757.7
F12_S05	shielded_dpp	low	P3	0	–	–	84.9	152.9	159.8
F12_S05	shielded_dpp	mid	P1	30	0.5667	0.0000	1903.8	1973.6	1976.1
F12_S05	shielded_dpp	mid	P2	30	0.8333	0.0000	822.2	890.1	892.5
F12_S05	shielded_dpp	mid	P3	30	1.0000	1.0000	81.1	146.8	152.3
F12_S05	shielded_dpp	high	P1	90	0.9222	0.0000	1914.0	1990.0	1992.6
F12_S05	shielded_dpp	high	P2	90	0.9667	0.0000	701.7	774.2	778.5
F12_S05	shielded_dpp	high	P3	90	1.0000	1.0000	80.0	140.8	147.1
F12_S06	fixed_local	low	P1	0	–	–	1932.5	2004.1	2017.5
F12_S06	fixed_local	low	P2	0	–	–	752.8	824.8	837.3
F12_S06	fixed_local	low	P3	0	–	–	90.0	164.6	171.9
F12_S06	fixed_local	mid	P1	0	–	–	1937.2	2042.4	2059.8
F12_S06	fixed_local	mid	P2	0	–	–	673.6	769.8	782.5
F12_S06	fixed_local	mid	P3	0	–	–	90.2	159.5	166.9
F12_S06	fixed_local	high	P1	0	–	–	1909.3	1956.4	1959.2
F12_S06	fixed_local	high	P2	0	–	–	785.0	826.4	829.8
F12_S06	fixed_local	high	P3	0	–	–	90.5	162.6	169.1
F12_S06	fixed_summary	low	P1	0	–	–	1785.0	1821.6	1825.8
F12_S06	fixed_summary	low	P2	0	–	–	614.8	652.1	654.9
F12_S06	fixed_summary	low	P3	0	–	–	88.8	157.6	164.1
F12_S06	fixed_summary	mid	P1	0	–	–	1889.4	1925.6	1929.1
F12_S06	fixed_summary	mid	P2	0	–	–	741.5	779.2	781.7
F12_S06	fixed_summary	mid	P3	0	–	–	85.5	156.6	163.6
F12_S06	fixed_summary	high	P1	0	–	–	1854.5	1988.9	2023.9
F12_S06	fixed_summary	high	P2	0	–	–	659.0	780.9	817.9
F12_S06	fixed_summary	high	P3	0	–	–	91.7	165.0	171.2
F12_S06	fixed_full	low	P1	90	0.0000	0.0000	2001.6	2059.9	2106.9
F12_S06	fixed_full	low	P2	90	0.0000	0.0000	694.5	809.4	850.0
F12_S06	fixed_full	low	P3	90	0.0000	0.0000	80.2	143.3	148.5
F12_S06	fixed_full	mid	P1	90	0.0000	0.0000	1890.5	2019.8	2023.1
F12_S06	fixed_full	mid	P2	90	0.0000	0.0000	652.4	715.4	735.8
F12_S06	fixed_full	mid	P3	90	0.0000	0.0000	77.1	141.6	146.7
F12_S06	fixed_full	high	P1	90	0.9667	0.0000	1966.0	2015.7	2038.6
F12_S06	fixed_full	high	P2	90	0.9111	0.0000	734.7	797.3	799.7
F12_S06	fixed_full	high	P3	90	1.0000	1.0000	80.4	144.3	150.8
F12_S06	independent_dpp	low	P1	22	0.0000	0.0000	2005.8	2040.9	2043.7
F12_S06	independent_dpp	low	P2	22	0.0000	0.0000	696.2	797.4	800.0
F12_S06	independent_dpp	low	P3	22	0.0000	0.0000	84.9	156.4	161.7
F12_S06	independent_dpp	mid	P1	45	0.0000	0.0000	1969.6	1996.1	2000.6
F12_S06	independent_dpp	mid	P2	45	0.0000	0.0000	903.6	932.0	934.8
F12_S06	independent_dpp	mid	P3	45	0.0000	0.0000	79.1	144.5	150.1
F12_S06	independent_dpp	high	P1	90	0.9222	0.0000	1912.8	2042.7	2046.9
F12_S06	independent_dpp	high	P2	90	0.9444	0.0000	730.8	856.4	859.5
F12_S06	independent_dpp	high	P3	90	1.0000	1.0000	73.9	132.8	138.6
F12_S06	central_dpp	low	P1	0	–	–	1945.7	2067.9	2070.5
F12_S06	central_dpp	low	P2	0	–	–	746.1	825.0	827.8
F12_S06	central_dpp	low	P3	0	–	–	84.7	151.8	158.0
F12_S06	central_dpp	mid	P1	30	0.9000	0.0000	2001.6	2059.9	2070.6
F12_S06	central_dpp	mid	P2	30	0.8667	0.0000	869.7	935.4	937.7
F12_S06	central_dpp	mid	P3	30	1.0000	1.0000	89.2	151.7	158.0
F12_S06	central_dpp	high	P1	90	0.9444	0.0000	2175.3	2214.4	2217.3
F12_S06	central_dpp	high	P2	90	0.9667	0.0000	1059.9	1102.3	1105.7
F12_S06	central_dpp	high	P3	90	1.0000	1.0000	79.4	154.0	160.2
F12_S06	shielded_dpp	low	P1	0	–	–	1933.6	1972.0	1976.6
F12_S06	shielded_dpp	low	P2	0	–	–	775.1	817.3	820.9
F12_S06	shielded_dpp	low	P3	0	–	–	89.9	160.3	166.5
F12_S06	shielded_dpp	mid	P1	30	0.8333	0.0000	1830.6	1960.8	1963.5
F12_S06	shielded_dpp	mid	P2	30	0.9000	0.0000	688.1	761.8	764.0
F12_S06	shielded_dpp	mid	P3	30	1.0000	1.0000	82.1	150.9	156.6
F12_S06	shielded_dpp	high	P1	90	0.9778	0.0000	1922.9	1970.4	1972.8
F12_S06	shielded_dpp	high	P2	90	0.9556	0.0000	698.8	740.1	744.4

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S06	shielded_dpp	high	P3	90	1.0000	1.0000	88.2	149.6	155.1
F12_S07	fixed_local	low	P1	0	–	–	1944.9	2011.2	2014.7
F12_S07	fixed_local	low	P2	0	–	–	730.1	809.6	812.8
F12_S07	fixed_local	low	P3	0	–	–	88.8	160.6	167.3
F12_S07	fixed_local	mid	P1	0	–	–	1868.9	1943.8	1948.2
F12_S07	fixed_local	mid	P2	0	–	–	629.0	699.1	702.5
F12_S07	fixed_local	mid	P3	0	–	–	93.9	164.7	171.4
F12_S07	fixed_local	high	P1	0	–	–	2100.5	2245.8	2249.1
F12_S07	fixed_local	high	P2	0	–	–	895.1	1020.1	1024.0
F12_S07	fixed_local	high	P3	0	–	–	88.5	159.7	166.7
F12_S07	fixed_summary	low	P1	0	–	–	1997.9	2132.3	2136.3
F12_S07	fixed_summary	low	P2	0	–	–	675.1	801.3	804.5
F12_S07	fixed_summary	low	P3	0	–	–	84.1	151.1	156.9
F12_S07	fixed_summary	mid	P1	0	–	–	1927.0	1978.8	1985.2
F12_S07	fixed_summary	mid	P2	0	–	–	612.5	680.9	684.0
F12_S07	fixed_summary	mid	P3	0	–	–	87.5	157.8	165.0
F12_S07	fixed_summary	high	P1	0	–	–	2181.8	2223.8	2227.5
F12_S07	fixed_summary	high	P2	0	–	–	613.1	751.0	755.5
F12_S07	fixed_summary	high	P3	0	–	–	84.2	154.1	161.1
F12_S07	fixed_full	low	P1	90	0.0000	0.0000	1825.8	1945.2	1947.3
F12_S07	fixed_full	low	P2	90	0.0000	0.0000	557.2	701.7	705.8
F12_S07	fixed_full	low	P3	90	0.0000	0.0000	81.1	143.6	149.2
F12_S07	fixed_full	mid	P1	90	0.0000	0.0000	2023.6	2274.3	2277.8
F12_S07	fixed_full	mid	P2	90	0.0000	0.0000	771.6	952.8	955.2
F12_S07	fixed_full	mid	P3	90	0.0000	0.0000	77.4	139.8	145.4
F12_S07	fixed_full	high	P1	90	0.9778	0.0000	1983.4	2025.4	2028.5
F12_S07	fixed_full	high	P2	90	0.9333	0.0000	793.0	850.9	872.5
F12_S07	fixed_full	high	P3	90	1.0000	1.0000	79.3	145.9	150.9
F12_S07	independent_dpp	low	P1	22	0.0000	0.0000	1909.6	2101.8	2103.8
F12_S07	independent_dpp	low	P2	22	0.0000	0.0000	717.8	874.3	876.8
F12_S07	independent_dpp	low	P3	22	0.0000	0.0000	83.9	149.3	155.4
F12_S07	independent_dpp	mid	P1	45	0.0000	0.0000	2049.3	2104.3	2107.5
F12_S07	independent_dpp	mid	P2	45	0.0000	0.0000	941.2	966.4	970.3
F12_S07	independent_dpp	mid	P3	45	0.0000	0.0000	85.9	151.4	156.4
F12_S07	independent_dpp	high	P1	90	0.9667	0.0000	1967.2	2170.5	2176.7
F12_S07	independent_dpp	high	P2	90	0.8889	0.0000	782.6	931.8	1045.7
F12_S07	independent_dpp	high	P3	90	0.9889	0.9889	81.2	328.2	333.1
F12_S07	central_dpp	low	P1	0	–	–	2036.6	2077.1	2082.3
F12_S07	central_dpp	low	P2	0	–	–	723.5	764.1	768.0
F12_S07	central_dpp	low	P3	0	–	–	88.3	156.0	161.8
F12_S07	central_dpp	mid	P1	30	0.8667	0.0000	2137.4	2178.3	2188.0
F12_S07	central_dpp	mid	P2	30	0.8667	0.0000	861.1	927.6	931.2
F12_S07	central_dpp	mid	P3	30	1.0000	1.0000	87.2	159.0	164.3
F12_S07	central_dpp	high	P1	90	0.9778	0.0000	2061.6	2089.0	2093.5
F12_S07	central_dpp	high	P2	90	0.9333	0.0000	864.5	941.0	944.8
F12_S07	central_dpp	high	P3	90	1.0000	1.0000	81.4	144.8	150.1
F12_S07	shielded_dpp	low	P1	0	–	–	1894.6	1993.4	1996.3
F12_S07	shielded_dpp	low	P2	0	–	–	628.4	734.7	738.1
F12_S07	shielded_dpp	low	P3	0	–	–	85.6	158.3	163.7
F12_S07	shielded_dpp	mid	P1	30	0.9333	0.0000	2068.2	2134.6	2137.8
F12_S07	shielded_dpp	mid	P2	30	0.9333	0.0000	716.9	752.4	755.3
F12_S07	shielded_dpp	mid	P3	30	1.0000	1.0000	79.0	149.3	154.3
F12_S07	shielded_dpp	high	P1	90	0.9889	0.0000	1891.0	1956.7	1959.9
F12_S07	shielded_dpp	high	P2	90	0.9556	0.0000	704.9	781.2	833.6
F12_S07	shielded_dpp	high	P3	90	1.0000	1.0000	79.4	144.9	151.5
F12_S08	fixed_local	low	P1	0	–	–	1957.4	2013.6	2016.1
F12_S08	fixed_local	low	P2	0	–	–	685.9	732.8	735.9
F12_S08	fixed_local	low	P3	0	–	–	87.0	158.9	165.4
F12_S08	fixed_local	mid	P1	0	–	–	1817.2	1937.7	1954.1
F12_S08	fixed_local	mid	P2	0	–	–	614.7	724.4	744.3
F12_S08	fixed_local	mid	P3	0	–	–	88.0	158.5	164.9
F12_S08	fixed_local	high	P1	0	–	–	1937.6	2017.9	2035.3
F12_S08	fixed_local	high	P2	0	–	–	683.8	776.3	784.7
F12_S08	fixed_local	high	P3	0	–	–	89.9	164.2	170.0
F12_S08	fixed_summary	low	P1	0	–	–	1999.1	2079.8	2083.7
F12_S08	fixed_summary	low	P2	0	–	–	735.8	778.9	786.1
F12_S08	fixed_summary	low	P3	0	–	–	88.0	158.4	164.5
F12_S08	fixed_summary	mid	P1	0	–	–	1992.5	2077.9	2082.7
F12_S08	fixed_summary	mid	P2	0	–	–	745.7	848.4	854.3
F12_S08	fixed_summary	mid	P3	0	–	–	89.3	158.2	164.7
F12_S08	fixed_summary	high	P1	0	–	–	2038.2	2116.8	2119.5
F12_S08	fixed_summary	high	P2	0	–	–	750.9	807.1	811.8
F12_S08	fixed_summary	high	P3	0	–	–	87.3	161.2	167.8
F12_S08	fixed_full	low	P1	90	0.0000	0.0000	2158.3	2203.4	2206.0
F12_S08	fixed_full	low	P2	90	0.0000	0.0000	978.1	1030.7	1034.2
F12_S08	fixed_full	low	P3	90	0.0000	0.0000	75.1	136.5	142.3
F12_S08	fixed_full	mid	P1	90	0.0000	0.0000	1956.9	2041.8	2044.2
F12_S08	fixed_full	mid	P2	90	0.0000	0.0000	904.9	1014.7	1018.5
F12_S08	fixed_full	mid	P3	90	0.0000	0.0000	80.4	142.8	147.6
F12_S08	fixed_full	high	P1	90	0.9333	0.0000	2053.0	2190.7	2193.1
F12_S08	fixed_full	high	P2	90	0.9556	0.0000	908.0	1033.1	1036.6
F12_S08	fixed_full	high	P3	90	1.0000	1.0000	80.5	142.0	146.8
F12_S08	independent_dpp	low	P1	22	0.0000	0.0000	1874.7	2010.2	2012.1

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S08	independent_dpp	low	P2	22	0.0000	0.0000	773.6	889.6	892.9
F12_S08	independent_dpp	low	P3	22	0.0000	0.0000	82.8	150.6	158.3
F12_S08	independent_dpp	mid	P1	45	0.0000	0.0000	1854.8	1911.5	1913.6
F12_S08	independent_dpp	mid	P2	45	0.0000	0.0000	664.3	721.6	724.3
F12_S08	independent_dpp	mid	P3	45	0.0000	0.0000	78.2	140.8	145.7
F12_S08	independent_dpp	high	P1	90	0.9222	0.0000	2146.9	2229.6	2231.4
F12_S08	independent_dpp	high	P2	90	0.9333	0.0000	995.3	1104.8	1107.4
F12_S08	independent_dpp	high	P3	90	1.0000	1.0000	75.3	141.2	146.0
F12_S08	central_dpp	low	P1	0	–	–	2023.5	2362.2	2365.6
F12_S08	central_dpp	low	P2	0	–	–	750.6	1083.7	1087.1
F12_S08	central_dpp	low	P3	0	–	–	83.0	154.3	161.6
F12_S08	central_dpp	mid	P1	30	0.9667	0.0000	2057.4	2105.0	2107.2
F12_S08	central_dpp	mid	P2	30	0.6667	0.0000	784.2	856.1	858.2
F12_S08	central_dpp	mid	P3	30	1.0000	1.0000	79.3	146.0	152.4
F12_S08	central_dpp	high	P1	90	0.9222	0.0000	2048.7	2112.0	2115.0
F12_S08	central_dpp	high	P2	90	0.9556	0.0000	814.7	873.6	877.0
F12_S08	central_dpp	high	P3	90	1.0000	1.0000	80.4	143.1	148.8
F12_S08	shielded_dpp	low	P1	0	–	–	1979.4	2072.9	2075.3
F12_S08	shielded_dpp	low	P2	0	–	–	746.4	815.6	819.0
F12_S08	shielded_dpp	low	P3	0	–	–	87.2	154.6	160.7
F12_S08	shielded_dpp	mid	P1	30	0.7333	0.0000	1895.4	1977.8	1980.2
F12_S08	shielded_dpp	mid	P2	30	0.7000	0.0000	754.9	834.4	837.4
F12_S08	shielded_dpp	mid	P3	30	1.0000	1.0000	84.4	149.1	155.6
F12_S08	shielded_dpp	high	P1	90	0.9556	0.0000	2070.2	2145.7	2150.4
F12_S08	shielded_dpp	high	P2	90	0.9556	0.0000	946.3	1008.9	1011.7
F12_S08	shielded_dpp	high	P3	90	1.0000	1.0000	72.2	135.7	140.7
F12_S09	fixed_local	low	P1	0	–	–	2062.2	2148.9	2153.1
F12_S09	fixed_local	low	P2	0	–	–	835.3	908.6	912.9
F12_S09	fixed_local	low	P3	0	–	–	91.8	166.5	174.5
F12_S09	fixed_local	mid	P1	0	–	–	2017.6	2068.9	2073.1
F12_S09	fixed_local	mid	P2	0	–	–	734.4	788.0	793.2
F12_S09	fixed_local	mid	P3	0	–	–	84.3	154.2	160.5
F12_S09	fixed_local	high	P1	0	–	–	2105.4	2162.6	2166.2
F12_S09	fixed_local	high	P2	0	–	–	862.7	933.5	937.1
F12_S09	fixed_local	high	P3	0	–	–	90.6	162.1	167.7
F12_S09	fixed_summary	low	P1	0	–	–	2035.6	2133.6	2137.1
F12_S09	fixed_summary	low	P2	0	–	–	740.9	828.5	832.8
F12_S09	fixed_summary	low	P3	0	–	–	91.4	166.4	172.2
F12_S09	fixed_summary	mid	P1	0	–	–	2067.9	2207.0	2209.7
F12_S09	fixed_summary	mid	P2	0	–	–	731.5	865.4	869.0
F12_S09	fixed_summary	mid	P3	0	–	–	89.9	163.7	169.8
F12_S09	fixed_summary	high	P1	0	–	–	1931.7	2076.8	2079.4
F12_S09	fixed_summary	high	P2	0	–	–	656.3	775.4	778.8
F12_S09	fixed_summary	high	P3	0	–	–	84.7	152.8	158.7
F12_S09	fixed_full	low	P1	90	0.0000	0.0000	2058.8	2117.3	2124.4
F12_S09	fixed_full	low	P2	90	0.0000	0.0000	756.7	835.5	843.0
F12_S09	fixed_full	low	P3	90	0.0000	0.0000	77.2	141.2	148.3
F12_S09	fixed_full	mid	P1	90	0.0000	0.0000	2005.3	2062.5	2089.6
F12_S09	fixed_full	mid	P2	90	0.0000	0.0000	877.8	931.8	935.3
F12_S09	fixed_full	mid	P3	90	0.0000	0.0000	77.9	136.7	142.7
F12_S09	fixed_full	high	P1	90	0.9111	0.0000	2041.2	2101.3	2106.6
F12_S09	fixed_full	high	P2	90	0.9556	0.0000	795.5	870.5	874.5
F12_S09	fixed_full	high	P3	90	1.0000	1.0000	77.1	141.8	147.0
F12_S09	independent_dpp	low	P1	22	0.0000	0.0000	2044.8	2140.4	2142.8
F12_S09	independent_dpp	low	P2	22	0.0000	0.0000	886.4	937.7	959.9
F12_S09	independent_dpp	low	P3	22	0.0000	0.0000	85.2	148.8	157.8
F12_S09	independent_dpp	mid	P1	45	0.0000	0.0000	2163.0	2195.9	2197.6
F12_S09	independent_dpp	mid	P2	45	0.0000	0.0000	1015.9	1045.1	1065.2
F12_S09	independent_dpp	mid	P3	45	0.0000	0.0000	78.9	140.5	145.3
F12_S09	independent_dpp	high	P1	90	0.9333	0.0000	2125.5	2172.7	2174.9
F12_S09	independent_dpp	high	P2	90	0.9333	0.0000	851.2	955.2	959.5
F12_S09	independent_dpp	high	P3	90	1.0000	1.0000	74.2	135.0	139.5
F12_S09	central_dpp	low	P1	0	–	–	1974.6	2062.7	2073.2
F12_S09	central_dpp	low	P2	0	–	–	700.8	778.4	783.7
F12_S09	central_dpp	low	P3	0	–	–	88.6	160.8	166.0
F12_S09	central_dpp	mid	P1	30	0.9000	0.0000	2105.6	2140.7	2143.5
F12_S09	central_dpp	mid	P2	30	0.7667	0.0000	957.1	996.2	1018.1
F12_S09	central_dpp	mid	P3	30	1.0000	1.0000	85.9	153.4	158.5
F12_S09	central_dpp	high	P1	90	0.9556	0.0000	2328.7	2368.8	2373.6
F12_S09	central_dpp	high	P2	90	0.9778	0.0000	997.1	1036.6	1039.8
F12_S09	central_dpp	high	P3	90	1.0000	1.0000	68.1	135.1	142.1
F12_S09	shielded_dpp	low	P1	0	–	–	1968.5	2040.6	2046.2
F12_S09	shielded_dpp	low	P2	0	–	–	790.4	868.6	870.5
F12_S09	shielded_dpp	low	P3	0	–	–	92.7	162.5	168.8
F12_S09	shielded_dpp	mid	P1	30	0.8667	0.0000	2063.5	2130.1	2140.9
F12_S09	shielded_dpp	mid	P2	30	0.9000	0.0000	834.2	885.0	888.3
F12_S09	shielded_dpp	mid	P3	30	1.0000	1.0000	80.4	145.5	150.6
F12_S09	shielded_dpp	high	P1	90	0.9444	0.0000	2087.2	2138.8	2140.9
F12_S09	shielded_dpp	high	P2	90	0.9444	0.0000	824.1	897.0	899.7
F12_S09	shielded_dpp	high	P3	90	1.0000	1.0000	73.9	137.8	143.0
F12_S10	fixed_local	low	P1	0	–	–	1963.5	2064.8	2068.2
F12_S10	fixed_local	low	P2	0	–	–	761.3	837.0	839.8
F12_S10	fixed_local	low	P3	0	–	–	91.5	166.4	173.2

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S10	fixed_local	mid	P1	0	–	–	2011.0	2050.3	2053.7
F12_S10	fixed_local	mid	P2	0	–	–	715.1	755.4	758.7
F12_S10	fixed_local	mid	P3	0	–	–	90.9	162.7	169.3
F12_S10	fixed_local	high	P1	0	–	–	1937.4	2000.6	2003.6
F12_S10	fixed_local	high	P2	0	–	–	668.6	723.4	726.7
F12_S10	fixed_local	high	P3	0	–	–	93.9	163.6	170.2
F12_S10	fixed_summary	low	P1	0	–	–	2013.5	2103.8	2108.3
F12_S10	fixed_summary	low	P2	0	–	–	823.8	913.5	917.0
F12_S10	fixed_summary	low	P3	0	–	–	84.6	153.6	160.5
F12_S10	fixed_summary	mid	P1	0	–	–	2057.6	2136.9	2140.5
F12_S10	fixed_summary	mid	P2	0	–	–	946.8	1014.3	1018.1
F12_S10	fixed_summary	mid	P3	0	–	–	84.4	155.9	163.6
F12_S10	fixed_summary	high	P1	0	–	–	1943.4	2129.7	2134.3
F12_S10	fixed_summary	high	P2	0	–	–	764.8	939.7	943.1
F12_S10	fixed_summary	high	P3	0	–	–	87.3	163.1	170.0
F12_S10	fixed_full	low	P1	90	0.0000	0.0000	1967.7	2037.9	2041.3
F12_S10	fixed_full	low	P2	90	0.0000	0.0000	858.5	951.5	953.8
F12_S10	fixed_full	low	P3	90	0.0000	0.0000	76.8	138.7	143.3
F12_S10	fixed_full	mid	P1	90	0.0000	0.0000	2146.3	2221.4	2223.5
F12_S10	fixed_full	mid	P2	90	0.0000	0.0000	847.3	890.1	894.4
F12_S10	fixed_full	mid	P3	90	0.0000	0.0000	74.8	132.3	136.9
F12_S10	fixed_full	high	P1	90	0.9778	0.0000	2165.6	2199.9	2204.4
F12_S10	fixed_full	high	P2	90	0.9667	0.0000	871.9	912.1	916.5
F12_S10	fixed_full	high	P3	90	1.0000	1.0000	81.3	142.9	147.6
F12_S10	independent_dpp	low	P1	22	0.0000	0.0000	1921.7	2011.9	2016.7
F12_S10	independent_dpp	low	P2	22	0.0000	0.0000	653.8	788.2	791.3
F12_S10	independent_dpp	low	P3	22	0.0000	0.0000	82.9	150.7	156.6
F12_S10	independent_dpp	mid	P1	45	0.0000	0.0000	1887.2	1966.4	1968.3
F12_S10	independent_dpp	mid	P2	45	0.0000	0.0000	773.0	838.0	840.0
F12_S10	independent_dpp	mid	P3	45	0.0000	0.0000	75.8	141.7	147.3
F12_S10	independent_dpp	high	P1	90	0.8556	0.0000	1965.1	2090.0	2093.2
F12_S10	independent_dpp	high	P2	90	0.9667	0.0000	882.8	981.7	984.1
F12_S10	independent_dpp	high	P3	90	1.0000	1.0000	76.0	142.0	147.0
F12_S10	central_dpp	low	P1	0	–	–	2041.7	2115.9	2119.5
F12_S10	central_dpp	low	P2	0	–	–	834.2	905.3	909.7
F12_S10	central_dpp	low	P3	0	–	–	89.3	164.3	169.0
F12_S10	central_dpp	mid	P1	30	0.9667	0.0000	2076.9	2105.3	2151.2
F12_S10	central_dpp	mid	P2	30	0.9000	0.0000	850.4	929.6	931.8
F12_S10	central_dpp	mid	P3	30	1.0000	1.0000	80.3	148.5	154.8
F12_S10	central_dpp	high	P1	90	0.9556	0.0000	2040.7	2102.1	2106.1
F12_S10	central_dpp	high	P2	90	0.9667	0.0000	905.9	940.3	944.7
F12_S10	central_dpp	high	P3	90	1.0000	1.0000	76.4	140.4	146.7
F12_S10	shielded_dpp	low	P1	0	–	–	2085.7	2128.5	2130.9
F12_S10	shielded_dpp	low	P2	0	–	–	938.2	977.5	981.4
F12_S10	shielded_dpp	low	P3	0	–	–	89.8	160.9	166.9
F12_S10	shielded_dpp	mid	P1	30	0.8000	0.0000	1980.5	2053.7	2058.1
F12_S10	shielded_dpp	mid	P2	30	0.7667	0.0000	729.0	827.8	830.2
F12_S10	shielded_dpp	mid	P3	30	1.0000	1.0000	86.9	154.8	160.6
F12_S10	shielded_dpp	high	P1	90	0.9222	0.0000	2020.4	2096.1	2123.6
F12_S10	shielded_dpp	high	P2	90	0.9778	0.0000	879.9	928.6	932.5
F12_S10	shielded_dpp	high	P3	90	1.0000	1.0000	80.1	143.5	149.0
F12_S11	fixed_local	low	P1	0	–	–	1923.5	1986.9	2001.2
F12_S11	fixed_local	low	P2	0	–	–	743.9	820.1	835.6
F12_S11	fixed_local	low	P3	0	–	–	89.1	161.4	168.1
F12_S11	fixed_local	mid	P1	0	–	–	2032.4	2109.7	2112.1
F12_S11	fixed_local	mid	P2	0	–	–	748.3	824.6	828.8
F12_S11	fixed_local	mid	P3	0	–	–	93.9	169.5	176.2
F12_S11	fixed_local	high	P1	0	–	–	2031.0	2105.3	2109.9
F12_S11	fixed_local	high	P2	0	–	–	736.1	814.3	818.7
F12_S11	fixed_local	high	P3	0	–	–	90.1	162.4	168.3
F12_S11	fixed_summary	low	P1	0	–	–	1928.4	1983.9	1990.1
F12_S11	fixed_summary	low	P2	0	–	–	639.4	714.9	725.7
F12_S11	fixed_summary	low	P3	0	–	–	82.4	153.2	159.2
F12_S11	fixed_summary	mid	P1	0	–	–	2087.9	2136.3	2140.6
F12_S11	fixed_summary	mid	P2	0	–	–	824.7	887.7	894.9
F12_S11	fixed_summary	mid	P3	0	–	–	86.8	157.3	162.5
F12_S11	fixed_summary	high	P1	0	–	–	1941.6	1992.7	1995.2
F12_S11	fixed_summary	high	P2	0	–	–	648.9	703.8	707.1
F12_S11	fixed_summary	high	P3	0	–	–	86.9	155.8	163.5
F12_S11	fixed_full	low	P1	90	0.0000	0.0000	1942.0	2002.2	2006.4
F12_S11	fixed_full	low	P2	90	0.0000	0.0000	724.1	806.1	808.5
F12_S11	fixed_full	low	P3	90	0.0000	0.0000	85.2	150.6	155.4
F12_S11	fixed_full	mid	P1	90	0.0000	0.0000	2070.3	2149.2	2152.6
F12_S11	fixed_full	mid	P2	90	0.0000	0.0000	938.8	980.4	983.0
F12_S11	fixed_full	mid	P3	90	0.0000	0.0000	79.7	142.6	148.2
F12_S11	fixed_full	high	P1	90	0.9111	0.0000	2113.8	2201.4	2203.3
F12_S11	fixed_full	high	P2	90	0.9333	0.0000	912.5	1038.8	1042.4
F12_S11	fixed_full	high	P3	90	1.0000	1.0000	76.8	135.8	141.6
F12_S11	independent_dpp	low	P1	22	0.0000	0.0000	1986.8	2019.7	2022.3
F12_S11	independent_dpp	low	P2	22	0.0000	0.0000	764.0	802.3	804.8
F12_S11	independent_dpp	low	P3	22	0.0000	0.0000	88.9	156.0	162.7
F12_S11	independent_dpp	mid	P1	45	0.0000	0.0000	2001.6	2087.2	2091.6
F12_S11	independent_dpp	mid	P2	45	0.0000	0.0000	824.6	904.2	906.9

Session	Method	Cap.	Dev.	n_c	Post	Nom.	Late p50	p95	p99
F12_S11	independent_dpp	mid	P3	45	0.0000	0.0000	78.9	141.5	146.5
F12_S11	independent_dpp	high	P1	90	0.9000	0.0000	2133.6	2182.2	2184.9
F12_S11	independent_dpp	high	P2	90	0.9444	0.0000	981.2	1046.9	1051.6
F12_S11	independent_dpp	high	P3	90	1.0000	1.0000	71.8	129.6	136.5
F12_S11	central_dpp	low	P1	0	–	–	2043.7	2116.0	2120.4
F12_S11	central_dpp	low	P2	0	–	–	783.2	865.3	868.8
F12_S11	central_dpp	low	P3	0	–	–	93.5	162.4	168.8
F12_S11	central_dpp	mid	P1	30	0.8000	0.0000	2106.5	2167.1	2169.4
F12_S11	central_dpp	mid	P2	30	0.8333	0.0000	934.4	991.7	995.4
F12_S11	central_dpp	mid	P3	30	1.0000	1.0000	80.0	146.7	151.1
F12_S11	central_dpp	high	P1	90	0.8889	0.0000	1923.1	1979.1	1982.6
F12_S11	central_dpp	high	P2	90	0.9444	0.0000	689.4	782.6	785.7
F12_S11	central_dpp	high	P3	90	1.0000	1.0000	75.7	135.1	140.8
F12_S11	shielded_dpp	low	P1	0	–	–	2014.5	2077.7	2080.4
F12_S11	shielded_dpp	low	P2	0	–	–	735.4	818.5	822.9
F12_S11	shielded_dpp	low	P3	0	–	–	87.8	158.6	165.2
F12_S11	shielded_dpp	mid	P1	30	0.8333	0.0000	1884.2	1947.6	1950.1
F12_S11	shielded_dpp	mid	P2	30	0.8667	0.0000	802.2	834.0	882.2
F12_S11	shielded_dpp	mid	P3	30	1.0000	1.0000	79.7	142.8	149.4
F12_S11	shielded_dpp	high	P1	90	0.9000	0.0000	1866.9	1926.8	1928.9
F12_S11	shielded_dpp	high	P2	90	0.9333	0.0000	651.9	721.0	724.2
F12_S11	shielded_dpp	high	P3	90	1.0000	1.0000	78.9	143.3	149.7
F12_S12	fixed_local	low	P1	0	–	–	2021.6	2060.8	2064.9
F12_S12	fixed_local	low	P2	0	–	–	851.3	891.4	896.0
F12_S12	fixed_local	low	P3	0	–	–	89.4	162.6	169.8
F12_S12	fixed_local	mid	P1	0	–	–	2065.4	2110.6	2113.4
F12_S12	fixed_local	mid	P2	0	–	–	880.8	918.2	921.2
F12_S12	fixed_local	mid	P3	0	–	–	95.8	167.4	173.3
F12_S12	fixed_local	high	P1	0	–	–	2028.6	2158.6	2162.3
F12_S12	fixed_local	high	P2	0	–	–	775.1	837.5	843.8
F12_S12	fixed_local	high	P3	0	–	–	92.3	164.3	171.5
F12_S12	fixed_summary	low	P1	0	–	–	1910.3	1985.2	1989.2
F12_S12	fixed_summary	low	P2	0	–	–	718.9	806.6	808.8
F12_S12	fixed_summary	low	P3	0	–	–	88.1	159.9	165.1
F12_S12	fixed_summary	mid	P1	0	–	–	1893.9	1959.5	1962.4
F12_S12	fixed_summary	mid	P2	0	–	–	613.9	675.0	678.6
F12_S12	fixed_summary	mid	P3	0	–	–	88.1	155.1	161.6
F12_S12	fixed_summary	high	P1	0	–	–	2217.8	2256.9	2260.5
F12_S12	fixed_summary	high	P2	0	–	–	1057.1	1098.5	1102.7
F12_S12	fixed_summary	high	P3	0	–	–	82.4	150.6	156.6
F12_S12	fixed_full	low	P1	90	0.0000	0.0000	2056.0	2127.3	2128.7
F12_S12	fixed_full	low	P2	90	0.0000	0.0000	831.5	884.5	889.1
F12_S12	fixed_full	low	P3	90	0.0000	0.0000	85.1	157.7	163.5
F12_S12	fixed_full	mid	P1	90	0.0000	0.0000	1859.9	1997.8	2002.3
F12_S12	fixed_full	mid	P2	90	0.0000	0.0000	697.3	841.4	844.7
F12_S12	fixed_full	mid	P3	90	0.0000	0.0000	83.0	145.3	151.3
F12_S12	fixed_full	high	P1	90	0.9222	0.0000	2070.3	2135.0	2139.0
F12_S12	fixed_full	high	P2	90	0.9444	0.0000	956.1	1001.6	1003.5
F12_S12	fixed_full	high	P3	90	1.0000	1.0000	80.0	138.6	145.2
F12_S12	independent_dpp	low	P1	22	0.0000	0.0000	1001.7	1038.7	1041.5
F12_S12	independent_dpp	low	P2	22	0.0000	0.0000	864.1	915.7	918.1
F12_S12	independent_dpp	low	P3	22	0.0000	0.0000	85.0	157.5	162.6
F12_S12	independent_dpp	mid	P1	45	0.0000	0.0000	2161.5	2193.0	2197.7
F12_S12	independent_dpp	mid	P2	45	0.0000	0.0000	860.3	896.0	897.8
F12_S12	independent_dpp	mid	P3	45	0.0000	0.0000	77.0	138.3	143.6
F12_S12	independent_dpp	high	P1	90	0.9444	0.0000	2004.9	2052.8	2056.1
F12_S12	independent_dpp	high	P2	90	0.9444	0.0000	778.7	819.1	822.4
F12_S12	independent_dpp	high	P3	90	1.0000	1.0000	74.8	135.8	141.5
F12_S12	central_dpp	low	P1	0	–	–	2072.6	2369.1	2372.7
F12_S12	central_dpp	low	P2	0	–	–	865.3	1186.7	1191.2
F12_S12	central_dpp	low	P3	0	–	–	86.1	161.1	167.6
F12_S12	central_dpp	mid	P1	30	0.5333	0.0000	1883.9	1935.5	1939.8
F12_S12	central_dpp	mid	P2	30	0.7667	0.0000	659.2	732.8	735.7
F12_S12	central_dpp	mid	P3	30	1.0000	1.0000	83.3	144.6	150.7
F12_S12	central_dpp	high	P1	90	0.9222	0.0000	2095.8	2187.1	2190.5
F12_S12	central_dpp	high	P2	90	0.9444	0.0000	846.9	919.8	922.5
F12_S12	central_dpp	high	P3	90	1.0000	1.0000	81.8	147.6	152.4
F12_S12	shielded_dpp	low	P1	0	–	–	1879.5	1960.7	1964.6
F12_S12	shielded_dpp	low	P2	0	–	–	695.3	756.2	759.7
F12_S12	shielded_dpp	low	P3	0	–	–	87.3	156.0	161.3
F12_S12	shielded_dpp	mid	P1	30	0.9000	0.0000	2041.9	2073.2	2075.5
F12_S12	shielded_dpp	mid	P2	30	0.9333	0.0000	803.7	835.2	837.5
F12_S12	shielded_dpp	mid	P3	30	1.0000	1.0000	83.1	147.1	152.7
F12_S12	shielded_dpp	high	P1	90	0.9444	0.0000	1962.8	2023.5	2026.7
F12_S12	shielded_dpp	high	P2	90	0.9444	0.0000	736.5	774.2	776.1
F12_S12	shielded_dpp	high	P3	90	1.0000	1.0000	74.7	136.9	142.8

VIII. RELEASE-LATENESS DIAGNOSTIC

Independent nominal clocks, the three-probe barrier, unequal first-iteration waiting, relative sleep, and no absolute catch-up are source/log supported. The specific OS/network reason for the recurring proposal-arrival order is UNKNOWN.

TABLE VI: Release-latency diagnostic from accepted formal timestamps (ms).

Device	Events	p50	p95	p99	Unproven residual
P1	19440	1972.34	2173.36	2326.13	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.
P2	19440	763.90	976.78	1080.30	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.
P3	19440	83.56	154.08	167.94	The exact OS/network reason that proposals repeatedly reached the barrier in P3/P2/P1 order, and residual per-session variation, are UNKNOWN.

IX. FINITE- n SCALING CONFORMANCE

This is discrete-event conformance, not a 32-probe hardware scalability experiment.

TABLE VII: All finite- n discrete-event conformance cells.

n	ρ	C (B)	k^*	Sim. k	Theory frac.	Sim. frac.	Match
3	1/3	131072	0	0	0.000000	0.000000	PASS
3	1/2	196608	1	1	0.333333	0.333333	PASS
3	2/3	262144	1	1	0.333333	0.333333	PASS
3	3/4	294912	2	2	0.666667	0.666667	PASS
3	1/1	393216	3	3	1.000000	1.000000	PASS
8	1/3	349525	1	1	0.125000	0.125000	PASS
8	1/2	524288	3	3	0.375000	0.375000	PASS
8	2/3	699050	4	4	0.500000	0.500000	PASS
8	3/4	786432	5	5	0.625000	0.625000	PASS
8	1/1	1048576	8	8	1.000000	1.000000	PASS
16	1/3	699050	3	3	0.187500	0.187500	PASS
16	1/2	1048576	6	6	0.375000	0.375000	PASS
16	2/3	1398101	9	9	0.562500	0.562500	PASS
16	3/4	1572864	11	11	0.687500	0.687500	PASS
16	1/1	2097152	16	16	1.000000	1.000000	PASS
32	1/3	1398101	7	7	0.218750	0.218750	PASS
32	1/2	2097152	13	13	0.406250	0.406250	PASS
32	2/3	2796202	19	19	0.593750	0.593750	PASS
32	3/4	3145728	22	22	0.687500	0.687500	PASS
32	1/1	4194304	32	32	1.000000	1.000000	PASS

X. COMMON-DENOMINATOR CONTROLLER COMPARISON

The main paper’s mid-capacity comparison divides on-time full-core results by all 270 generated measured events. It is derived directly from the frozen phase-sensitivity CSV rather than from conditional coverage alone. The coordinated policy

TABLE VIII
MID-CAPACITY COMMON-DENOMINATOR SOFTWARE SENSITIVITY.

Controller	Full req.	Timely/gen.	Cond.	MB
C-QPG	0.333	0.3333	1.000	14.746
Desync. I-QPG	0.667	0.3352	0.503	23.593
Sync. I-QPG	0.500	0.0000	0.000	17.695

uses 37.5% fewer application bytes than the phase- randomized independent policy (14.7456 versus 23.59296 MB) while producing 0.3333 versus 0.3352 timely full results per generated event.

XI. INDEPENDENCE APPROXIMATION

At mid capacity, an independent Bernoulli request model with $p = 2/3$ gives $P(J = 1) = 6/27$, $P(J = 2) = 12/27$, and $P(J = 3) = 8/27$. If a $2B$ service budget finishes all requests for $J \leq 2$ and none for $J = 3$, the expected conditional timely fraction is $5/9 \approx 0.5556$ and the common timely-full fraction is $10/27 \approx 0.3704$. The phase-randomized simulator gives 0.5028 and 0.3352, respectively. The Bernoulli calculation is an independence approximation, not a proved upper bound.

XII. ROTATING-FRONTIER SOFTWARE CONFORMANCE

The cyclic construction uses the same 1-ms integer service accounting as the phase study. It admits exactly $K = \lfloor C/B \rfloor$ full requests per round and zero payload from the other probes. There are zero simulated shared-cap violations. This discrete-event

TABLE IX
IDEALIZED ROTATING-FRONTIER CONFORMANCE.

Capacity	K	Full fraction	Timely full / generated	Fair period
Low	1	0.3333	0.3333	3
Mid	2	0.6667	0.6667	3
High	3	1.0000	1.0000	1

check was completed before the separately frozen Raspberry Pi extension and is retained only as a conformance check on the cyclic byte-allocation rule; the physical results are reported separately below.

XIII. CRFA HARDWARE FRONTIER EXTENSION

The separately frozen CRFA campaign tests the same cyclic construction on the three physical probes without changing the primary A–F campaign. Twelve matched session/workload identities are executed at each capacity, yielding 36 accepted arms. Across 55 indexed attempts, 19 attempts are excluded for predefined infrastructure failures; all excluded attempts are retained in the attempt index, timing outcomes are never acceptance criteria, and partial attempts are not stitched into accepted arms. The accepted extension contains 9,720 measured occurrences and 10,620 reconstructed shared-service rounds. All frozen event-identity, action/payload, final-drain, raw-reconstruction, and shared-cap gates pass, with zero shared-cap violations.

TABLE X
CRFA PHYSICAL ATTAINMENT OF THE UNRESTRICTED FULL-EVIDENCE BYTE FRONTIER. THE C-QPG COLUMN USES THE SESSION-MATCHED PRIMARY HARDWARE RESULT; PAIRED INTERVALS ARE DESCRIPTIVE 95% SESSION-BOOTSTRAP INTERVALS.

Capacity	Byte frontier	C-QPG	CRFA	Attainment	Δ CRFA–C-QPG (95% CI)
Low	0.3333	0.0000	0.3056	0.9167	+0.3056 [+0.3015, +0.3093]
Mid	0.6667	0.2938	0.6327	0.9491	+0.3389 [+0.3306, +0.3478]
High	1.0000	0.9698	0.9660	0.9660	-0.0037 [-0.0133, +0.0046]

At low, mid, and high capacity, CRFA produces 0.3056, 0.6327, and 0.9660 timely full-core results per generated event, corresponding to 91.7%, 94.9%, and 96.6% of the byte frontier. Every selected full-evidence result is eventually returned; the residual to the byte frontier is therefore post-release timing loss rather than loss of selected evidence. At mid capacity, CRFA produces 2,050 timely full-core results from 3,240 generated occurrences, compared with 952 for session-matched C-QPG. The paired mean difference is +0.3389 with 95% session-bootstrap interval [0.3306, 0.3478]. At high capacity, where universal coverage no longer costs a full-evidence slot, CRFA–C-QPG is -0.0037 with interval $[-0.0133, 0.0046]$.

These results are not a universal dominance claim. CRFA assigns H-NONE to non-selected probes and therefore exchanges same-round remote coverage for more full-evidence opportunities. C-QPG instead preserves at least summary evidence for every probe in every round.

XIV. CRFA PER-DEVICE ADMISSION FAIRNESS

The cyclic construction equalizes full-evidence opportunity counts exactly, while the physical post-release timing of those opportunities remains device dependent. Across the 12 accepted sessions, each probe receives 360, 720, and 1,080 full-evidence

TABLE XI
CRFA PER-DEVICE ADMISSION OPPORTUNITY AND POST-RELEASE TIMING.

Capacity	Probe	Full opps.	Timely full	Cond. timely	Max sel. interval
Low	P1	360	314	0.8722	3
Low	P2	360	316	0.8778	3
Low	P3	360	360	1.0000	3
Mid	P1	720	659	0.9153	2
Mid	P2	720	671	0.9319	2
Mid	P3	720	720	1.0000	2
High	P1	1080	1021	0.9454	1
High	P2	1080	1030	0.9537	1
High	P3	1080	1079	0.9991	1

opportunities at low, mid, and high capacity, respectively. The maximum inter-selection interval is three, two, and one measured rounds. At mid capacity, conditional post-release on-time fractions for selected full events are 0.9153, 0.9319, and 1.0000 for P1, P2, and P3. Thus deterministic admission fairness does not imply identical physical timing.

XV. CRFA PER-DEVICE TIMING RECONSTRUCTION

TABLE XII
CRFA SELECTED-FULL TIMING BY DEVICE. NOMINAL ON-TIME REFERS TO THE SCHEDULE-INCLUSIVE ANCHOR; POST-RELEASE ON-TIME USES ACTUAL EVENT CREATION.

Capacity	Probe	Post timely	Nom. timely	Release p50 (ms)	Post p50 (ms)	Post p95 (ms)
Low	P1	0.8722	0.0000	2182.8	938.1	1042.3
Low	P2	0.8778	0.0000	1077.9	925.9	1028.7
Low	P3	1.0000	1.0000	82.2	245.4	340.6
Mid	P1	0.9153	0.0000	2142.1	928.2	1011.2
Mid	P2	0.9319	0.0000	903.5	886.0	1012.5
Mid	P3	1.0000	1.0000	78.4	238.4	338.9
High	P1	0.9454	0.0000	5079.6	900.2	1000.9
High	P2	0.9537	0.0000	3621.6	879.9	998.5
High	P3	0.9991	0.9991	77.0	238.8	333.0

P1 and P2 have zero nominal-anchor timely full results at all three capacities in the CRFA extension, whereas P3 is essentially fully nominally on time. The aggregate CRFA nominal timely-full fractions on the common generated-event denominator are 0.1111, 0.2222, and 0.3330 at low, mid, and high capacity, versus post-release values of 0.3056, 0.6327, and 0.9660. This reproduces the schedule-anchor asymmetry seen in the primary campaign and reinforces the scope of the primary 1-s quantity as a post-release service SLO.

XVI. CRFA PER-SESSION RESULTS

TABLE XIII
CRFA SESSION-LEVEL TIMELY FULL-CORE OUTPUT AND FRONTIER ATTAINMENT.

Session	Low timely	Low η	Mid timely	Mid η	High timely	High η
F12_S01	0.3000	0.9000	0.6259	0.9389	0.9704	0.9704
F12_S02	0.3000	0.9000	0.6333	0.9500	0.9519	0.9519
F12_S03	0.3074	0.9222	0.6111	0.9167	0.9481	0.9481
F12_S04	0.3074	0.9222	0.6296	0.9444	0.9704	0.9704
F12_S05	0.3111	0.9333	0.6259	0.9389	0.9667	0.9667
F12_S06	0.2926	0.8778	0.6333	0.9500	0.9852	0.9852
F12_S07	0.3037	0.9111	0.6370	0.9556	0.9667	0.9667
F12_S08	0.3074	0.9222	0.6407	0.9611	0.9519	0.9519
F12_S09	0.3111	0.9333	0.6370	0.9556	0.9815	0.9815
F12_S10	0.3111	0.9333	0.6519	0.9778	0.9778	0.9778
F12_S11	0.3185	0.9556	0.6407	0.9611	0.9667	0.9667
F12_S12	0.2963	0.8889	0.6259	0.9389	0.9556	0.9556

XVII. HARDWARE TIMING ANCHOR SUMMARY

TABLE XIV
ACCEPTED HARDWARE CORE-REQUIRED TIMING BY PROBE.

Probe	Post	Nominal	Lateness p50 (ms)
P1	0.583	0.000	1972.3
P2	0.588	0.000	763.9
P3	0.630	0.630	83.6

P1 and P2 have zero nominal-anchor on-time coverage when all accepted core-required occurrences are pooled by device. P1’s median release offset is about 1.97 s and therefore exceeds the 1-s service horizon before post-release service time is added. The exact OS/network cause of the recurring proposal arrival order remains unknown.

XVIII. ACCEPTED AND EXCLUDED ATTEMPTS

For the primary A–F campaign, accepted attempts are S01-a01, S02-a01, S03-a02, S04-a02, S05-a01, S06-a01, S07-a02, and S08–S12-a01. S03-a01, S04-a01, and S07-a01 are excluded infrastructure attempts and are not stitched into accepted sessions.

The CRFA extension indexes 55 attempts for 36 planned arms. Thirty-six arms are accepted and 19 attempts are excluded under the frozen infrastructure rules, predominantly SSH transport failures plus a pre-arm hardware-gate failure. The accepted and excluded lineage is preserved in the extension attempt index. Timing or favorable scientific outcomes are not retry criteria.

XIX. DETECTOR AND REPLAY LIMITATIONS

The workload is frozen CICIOT2023-derived feature-vector replay, not packet replay or an uncontrolled deployment. The dataset is cited in the main paper. The formal occurrence set excludes Recon-PortScan, and exact feature-vector overlap exists across frozen partitions. Perfect correctness on the formal occurrences is therefore a controlled diagnostic rather than an IDS-superiority result.

XX. RAW AUDIT AND OFFLINE-ANALYSIS VERDICTS

The primary full-raw audit rebuilt 216/216 accepted arms and 58,320 measured events with zero raw-to-metric mismatch and zero shared-cap violation across 63,720 service rounds. Its historical verdict remains `PASS_WITH_ISSUES` because the raw bundle lacks direct artifacts for the S03-a01 SSD incident and the S07 pre-hardware launcher abort; the CRFA extension does not alter that disclosure.

The separately archived CRFA evidence reconstructs 36/36 accepted arms, 9,720 measured occurrences, and 10,620 shared-service rounds. Its frozen identity, action/payload, final-drain, reconstruction, and cap gates pass, and no shared-cap violation is observed. Across the two hardware campaigns this corresponds to 252 accepted arms, 68,040 measured occurrences, and 74,340 reconstructed shared-service rounds, with zero observed shared-cap violations.

The read-only offline analyses contain 432 queue-weight rows, 240 phase- sensitivity rows, 648 primary schedule-anchor rows, and 20 multi- n scaling rows. The rotating software-conformance table remains a separate discrete- event check and is not relabelled as hardware evidence.

XXI. ARTIFACT ACCESS

Machine-readable CSV/JSON sources, including the CRFA arm/capacity/session summaries, occurrence ledger, service audit, fairness table, timing reconstruction, matched comparisons, attempt index, number provenance, claim audit, build report, validation report, and SHA-256 manifest accompany this supplementary PDF. The CRFA primary archive and integrity-checked mirror reside on the same user-authorized external physical volume and therefore do not constitute independent physical backups. Third-party dataset redistribution rights remain outside the scope of this package.

XXII. NOTATION INDEX

The main paper defines symbols where they enter the model. This index collects them by role; quantities with units of bytes are payload or service accounting quantities. The summary size S and cumulative service $S(T)$ are distinct despite sharing the letter.

TABLE XV: Notation index for the main model and supplementary planning extensions.

Symbol	Meaning and units
$n; i, t$	Probe count; probe index and coordination-round index.
$m, k; r, f$	Remotely represented/full counts; normalized fractions $r = m/n, f = k/n$.
$B, S; C(t), C$	Full/summary payload bytes; varying or constant shared bytes per round.
ρ, σ	Normalized capacity $C/(nB)$ and summary/full ratio S/B .
$k^*(C, m)$	Maximum full count while representing a fixed number m of probes.
$K_{\text{full}}^{\max}, K$	Unrestricted maximum full count; K is shorthand for this count.
$K_{\text{all}}^{\max}, \Pi_{\text{cov}}$	Maximum full count with all probes represented; full-slot coverage price.
$a_i(t); b(a), u(a)$	Admitted action; payload in bytes and frozen utility score.
$Q_i(t), Q_{\Sigma}(T)$	Per-probe and aggregate queued bytes.
$g_i(t), S(T)$	Per-probe service grant; cumulative aggregate bytes served through T .
$v, \psi_i(a, t), q_a$	Utility weight, queue-penalized score, and positive-payload queue threshold.
A_t, D	Aggregate admitted bytes in round t ; common relative deadline in rounds.
$\mathcal{F}_{\text{same}}, \mathcal{F}_D, \mathcal{F}_{\text{rate}}$	Feasible sets for same-round, finite-deadline, and long-run load conditions.
p, p^*	Full-request fraction and its conditional saturated-regime value. In the independent-request planning model, p is the Bernoulli request probability.
$\mathcal{S}_t, T_{\text{fair}}$	Rotating selected set and cycle period $n/\gcd(n, K)$.
$L_{\text{post}}, L_{\text{nom}}$	Result-return latency from actual creation or planned release, in seconds.
$t_{\text{created}}, t_{\text{scheduled}}, t_{\text{return}}$	Event creation, scheduled-release, and result-return timestamps.
Δ	Descriptive paired CRFA–C-QPG difference in timely-full/generated output.
$C_{\min}(k)$	Minimum byte budget for k full uploads while preserving all n summaries as a baseline.
$J, h(C), J_{\text{adm}}$	Random full-request count, summary-preserving full capacity, and admitted full count.
$P_{\text{deg}}, \epsilon, q_{\epsilon}, C_{\epsilon}$	Probability of any downgrade, its target, request-count quantile, and planning budget.
$\mathcal{M}, \mathcal{F}$	Fixed remotely represented set and its full-upgrade subset.
S_i, B_i, d_i, R	Per-probe payload bytes, upgrade cost $d_i = B_i - S_i$, and residual budget.
$w_i, x_i, k_{\text{het}}^*$	Upgrade value, binary choice, and maximum count for equal-valued heterogeneous upgrades.

XXIII. EXTENDED MECHANISM TRACE

Figure 1 preserves the complete archived 16-round frozen $v = 1$ mid-capacity replay illustration. The main paper uses only its first eight rounds to explain the period-two mechanism; the separate weight sweep addresses parameter robustness. No hardware traces, replay outcomes, or summary data were regenerated for this presentation change.

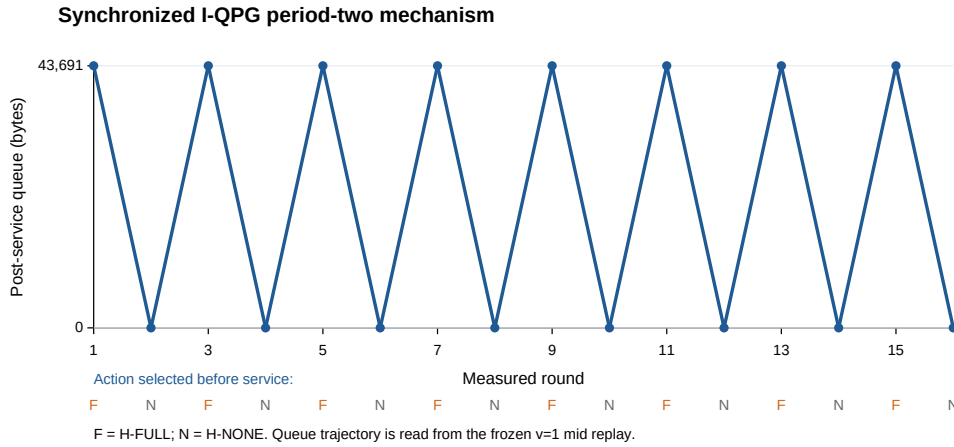


Fig. 1. Full archived I-QPG period-two illustration. F denotes H-FULL and N denotes H-NONE; values and action order are unchanged from the integrated submission candidate.